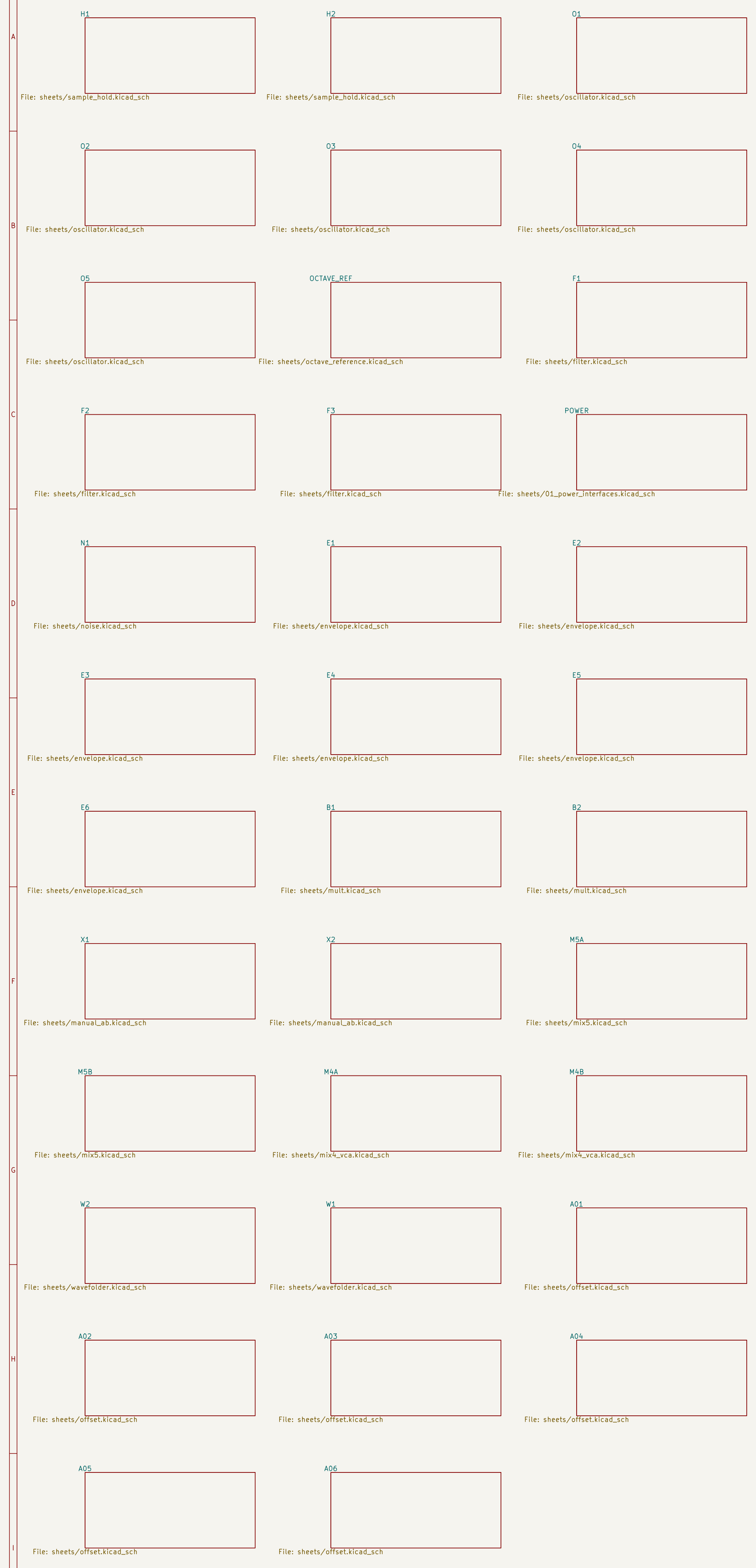
H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope.kicad_sch
E3
File: sheets/envelope.kicad_sch
E4
File: sheets/envelope.kicad_sch
E5
File: sheets/envelope.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

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File: zudo-osc-hole-field.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
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Sheet: /H1/
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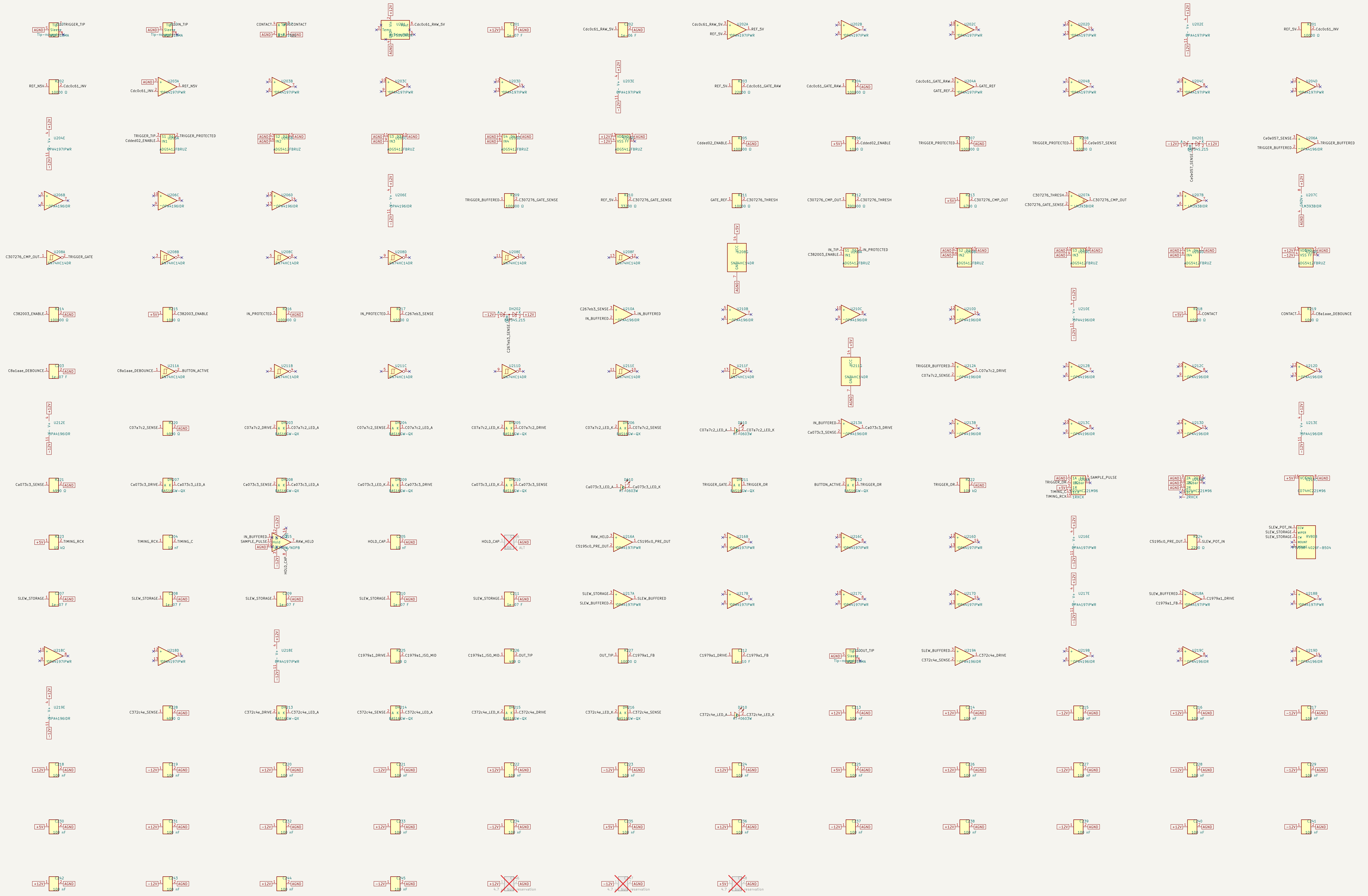
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/36



TRIGGER_TIP
DIN_TIP
CONTACT
U201
Cdc0c61_RAW_5V
C201
1e-07 F
C202
1e-06 F
U202A
REF_5V
OPA4197IPWR
U202B
U202C
U202D
U202E
R201
Cdc0c61_INV
10000 Ω
R202
REF_N5V
U203A
U203B
U203C
U203D
U203E
R203
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22000 Ω
R204
100000 Ω
U204A
GATE_REF
U204B
U204C
U204D
U204E
U205
Cdded02_ENABLE
TRIGGER_PROTECTED
ADG5412FBRUZ
R205
R206
R207
R208
Ce0e057_SENSE
DH201
BAT54S,215
U206A
TRIGGER_BUFFERED
OPA4196IDR
U206B
U206C
U206D
U206E
R209
C307276_GATE_SENSE
R210
R211
C307276_THRESH
R212
C307276_CMP_OUT
R213
4700 Ω
U207A
LM393BIDR
U207B
U207C
U208A
TRIGGER_GATE
SN74HC14DR
U208B
U208C
U208D
U208E
U208F
U208G
U209
IN_TIP
IN_PROTECTED
C382003_ENABLE
R214
R215
R216
R217
C267eb3_SENSE
DH202
U210A
IN_BUFFERED
U210B
U210C
U210D
U210E
R218
R219
C8a1aae_DEBOUNCE
C203
U211A
BUTTON_ACTIVE
U211B
U211C
U211D
U211E
U211F
U211G
U212A
C07a7c2_DRIVE
C07a7c2_SENSE
U212B
U212C
U212D
U212E
R220
D203
C07a7c2_LED_A
BAS16GW-QX
D204
D205
C07a7c2_LED_K
D206
U213A
Ca073c3_SENSE
Ca073c3_DRIVE
U213B
U213C
U213D
U213E
R221
D207
Ca073c3_LED_A
D208
D209
Ca073c3_LED_K
D211
TRIGGER_OR
D212
R222
10 kΩ
U214
SAMPLE_PULSE
TIMING_CX
TIMING_RCX
CD74HC221M96
R223
TIMING_RCX
C204
TIMING_C
U215
SAMPLE_PULSE
RAW_HELD
HOLD_CAP
C205
10 nF
C206
U216A
C5195c0_PRE_OUT
U216B
U216C
U216D
U216E
R224
SLEW_POT_IN
2200 Ω
RV801
SLEW_STORAGE
C207
C208
C209
C210
C211
U217A
SLEW_BUFFERED
U217B
U217C
U217D
U217E
U218A
C1979a1_DRIVE
C1979a1_FB
U218B
U218C
U218D
U218E
R225
C1979a1_ISO_MID
499 Ω
R226
OUT_TIP
R227
C212
1e-10 F
U219A
C372c4e_SENSE
C372c4e_DRIVE
U219B
U219C
U219D
U219E
R228
D213
C372c4e_LED_A
D214
D215
C372c4e_LED_K
D216
C213
C214
C215
C216
C217
100 nF
C218
C219
C220
C221
C222
C223
C224
C225
C226
C227
C228
C229
C230
C231
C232
C233
C234
C235
C236
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C238
C239
C240
C241
C242
C243
C244
C245
+12V
-12V
+5V
AGND

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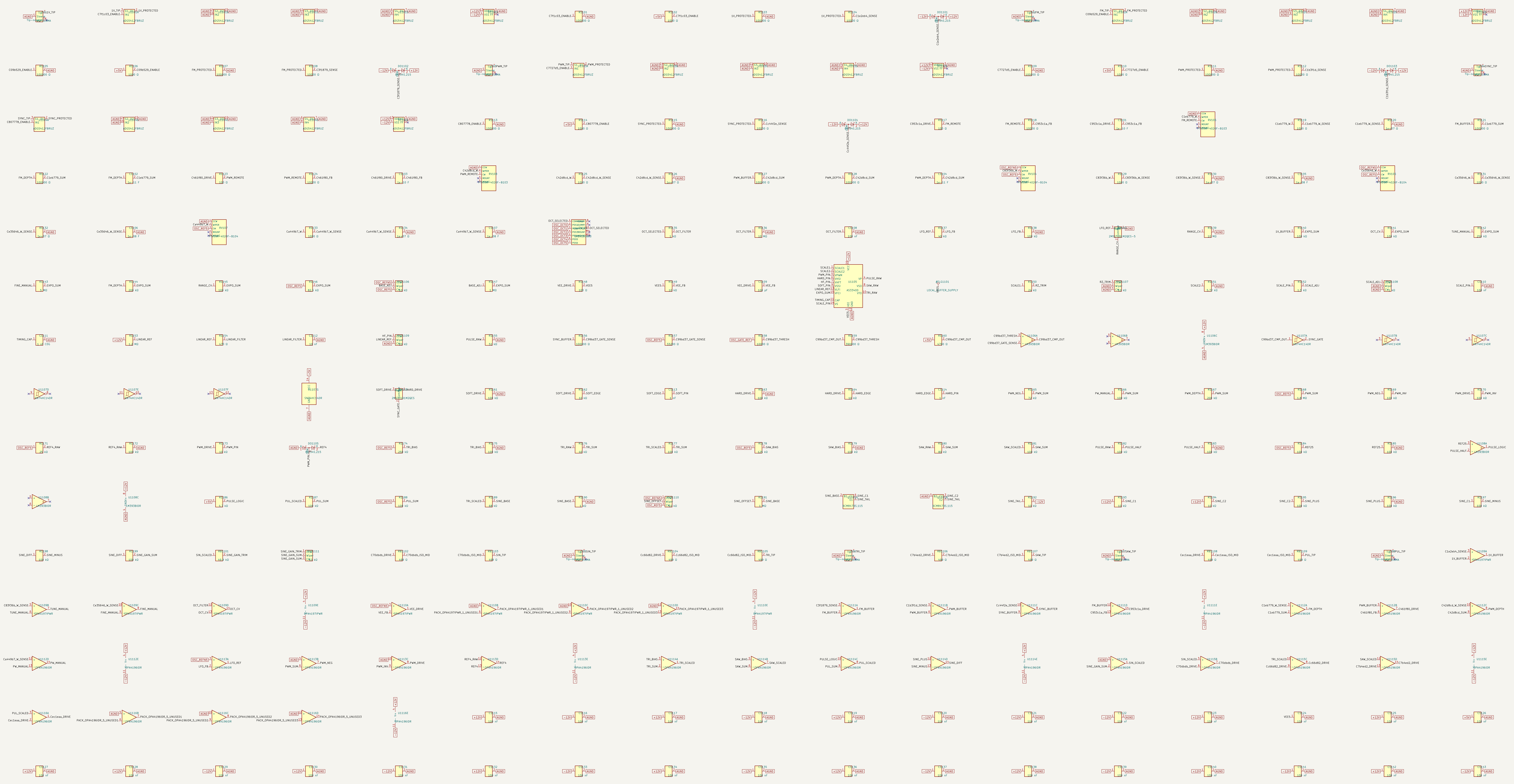
KiCad E.D.A. 10.0.6 | Id: 3/36



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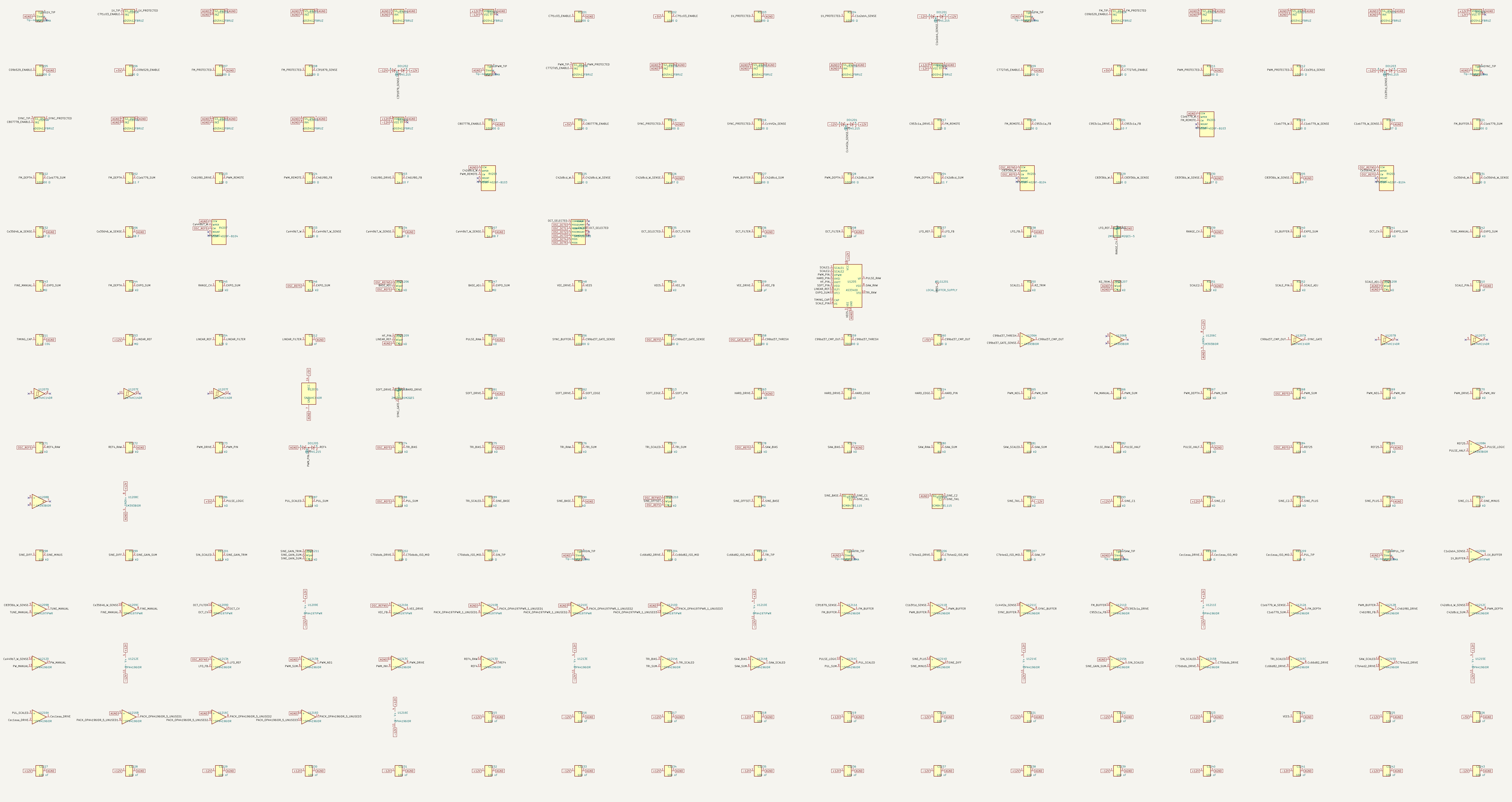
KiCad E.D.A. 10.0.6 | Id: 4/36



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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 5/36

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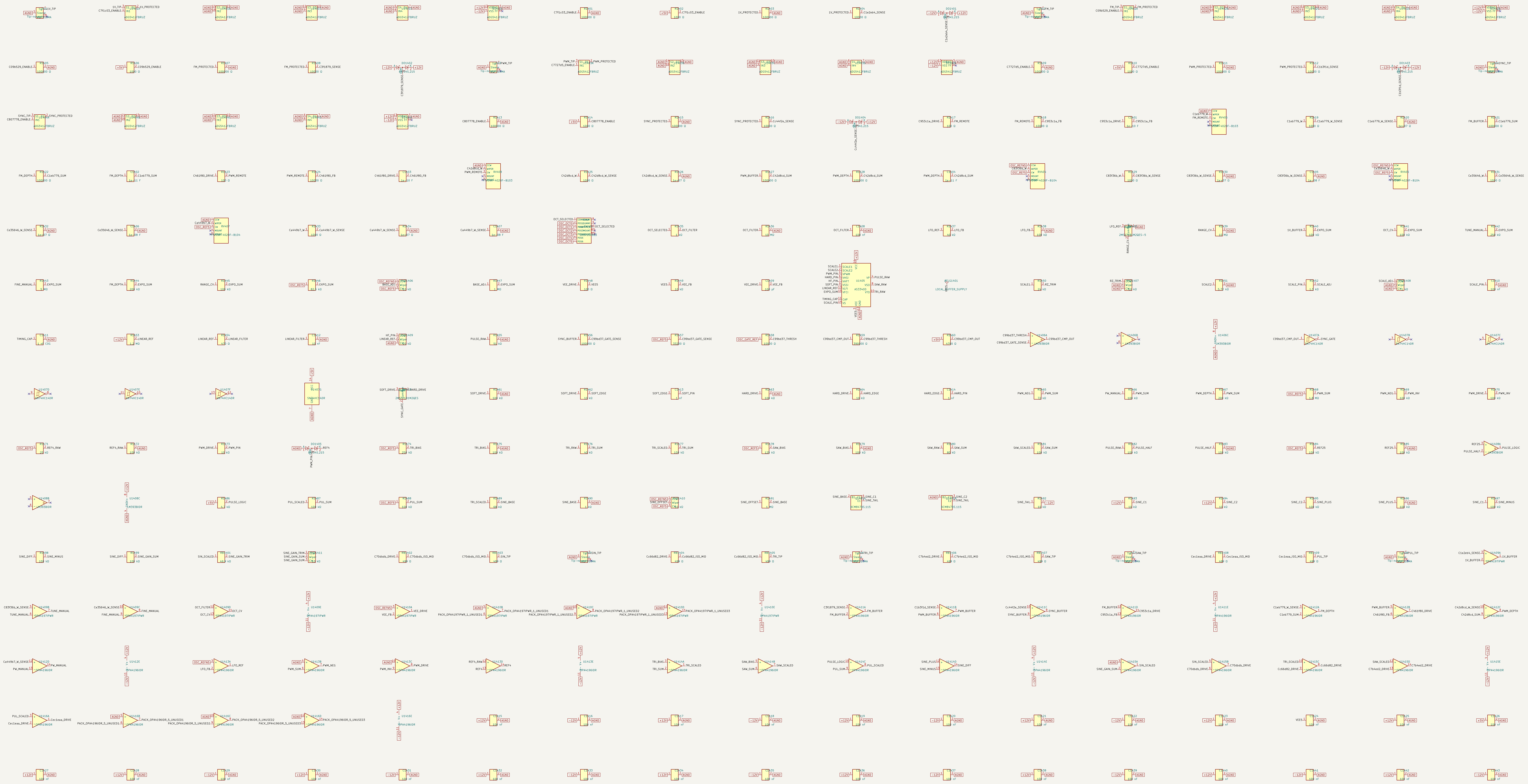
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/36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Sleeve
Tip-normal
AGND
1V_TIP
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U1401A
1V_PROTECTED
ADG5412FBRUZ
U1401B
U1401C
U1401D
C09b529_ENABLE
R1405
100000 Ω
+5V
R1406
1000 Ω
FM_PROTECTED
R1407
R1408
10000 Ω
C3f1879_SENSE
-12V
DO1402
+12V
BAT54S,215
SYNC_TIP
C807778_ENABLE
U1404A
SYNC_PROTECTED
U1404B
U1404C
U1404D
U1404E
FM_DEPTH
R1422
C1eb779_SUM
C1402
1e-11 F
C461f80_DRIVE
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100 Ω
PWM_REMOTE
R1424
C461f80_FB
C1403
1e-10 F

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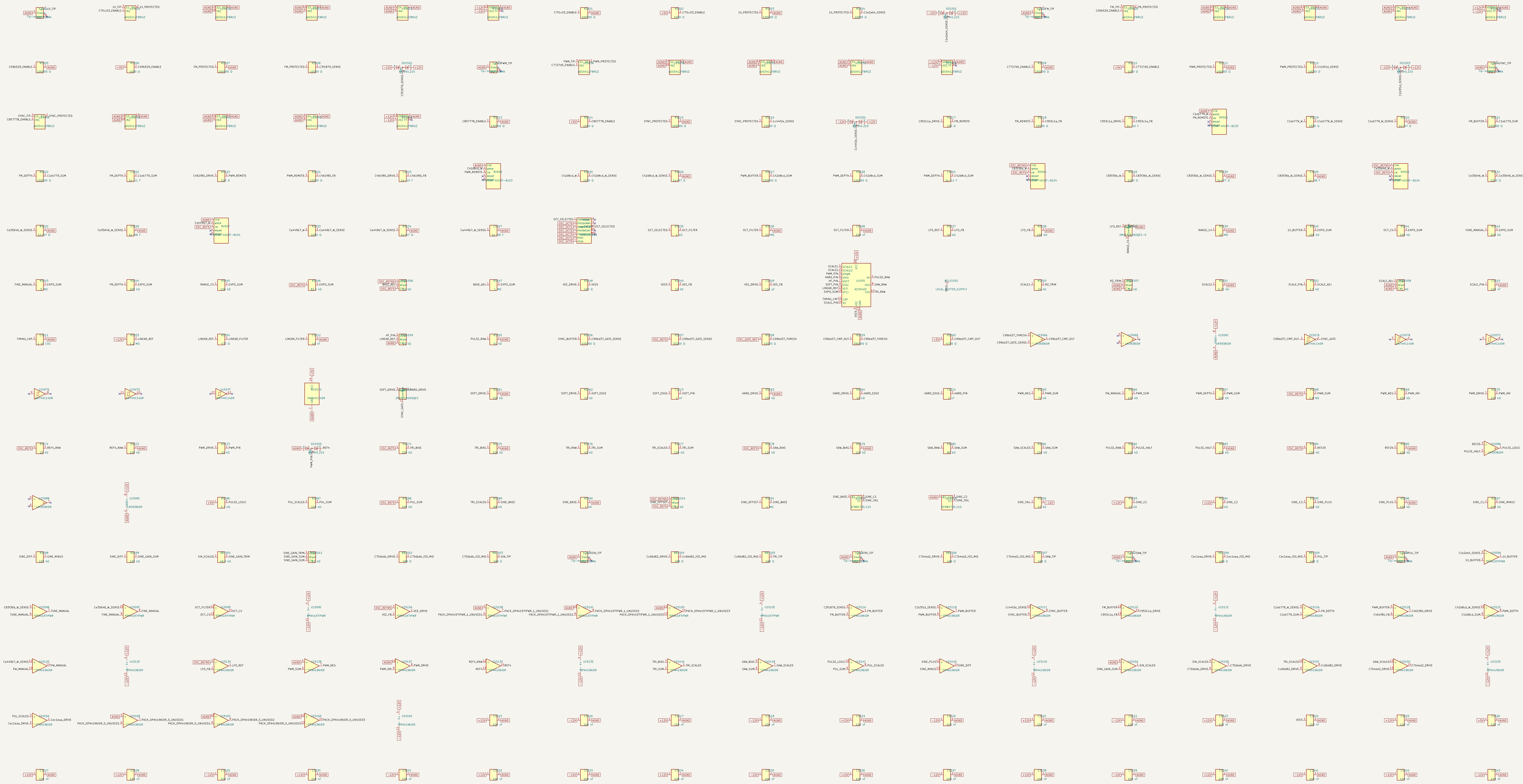
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 7/36



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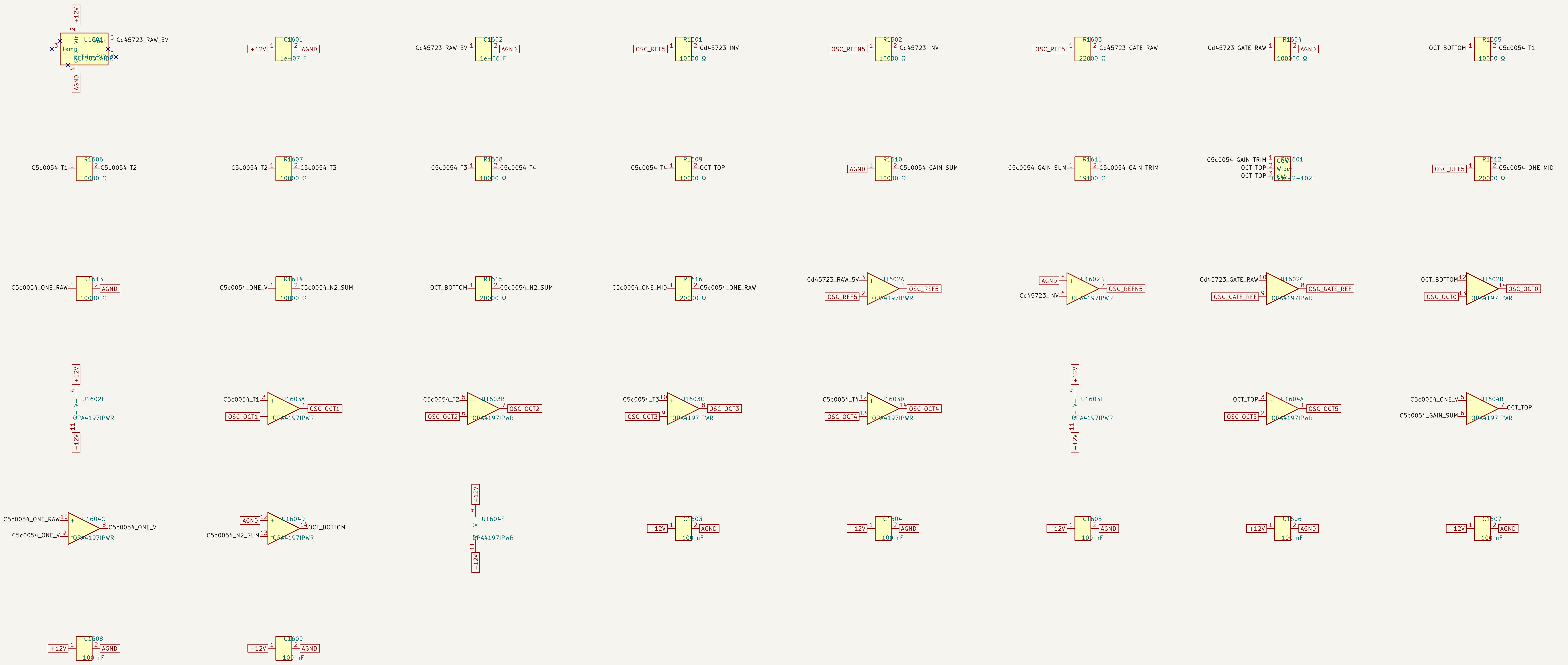
KiCad E.D.A. 10.0.6		Id: 8/36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Vin
Vout
Temp
Trim/NR
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AGND
C1601
+12V
AGND
1e-07 F
C1602
Cd45723_RAW_5V
AGND
1e-06 F
R1601
OSC_REF5
Cd45723_INV
10000 Ω
R1602
OSC_REFN5
Cd45723_INV
10000 Ω
R1603
OSC_REF5
Cd45723_GATE_RAW
22000 Ω
R1604
Cd45723_GATE_RAW
AGND
100000 Ω
R1605
OCT_BOTTOM
C5c0054_T1
10000 Ω
R1606
C5c0054_T1
C5c0054_T2
10000 Ω
R1607
C5c0054_T2
C5c0054_T3
10000 Ω
R1608
C5c0054_T3
C5c0054_T4
10000 Ω
R1609
C5c0054_T4
OCT_TOP
10000 Ω
R1610
AGND
C5c0054_GAIN_SUM
10000 Ω
R1611
C5c0054_GAIN_SUM
C5c0054_GAIN_TRIM
19100 Ω
RV1601
C5c0054_GAIN_TRIM
OCT_TOP
OCT_TOP
CCW
Wiper
CW
TC33X-2-102E
R1612
OSC_REF5
C5c0054_ONE_MID
20000 Ω
R1613
C5c0054_ONE_RAW
AGND
10000 Ω
R1614
C5c0054_ONE_V
C5c0054_N2_SUM
10000 Ω
R1615
OCT_BOTTOM
C5c0054_N2_SUM
20000 Ω
R1616
C5c0054_ONE_MID
C5c0054_ONE_RAW
20000 Ω
U1602A
Cd45723_RAW_5V
OSC_REF5
OSC_REF5
OPA4197IPWR
U1602B
AGND
Cd45723_INV
OSC_REFN5
OPA4197IPWR
U1602C
Cd45723_GATE_RAW
OSC_GATE_REF
OSC_GATE_REF
OPA4197IPWR
U1602D
OCT_BOTTOM
OSC_OCT0
OSC_OCT0
OPA4197IPWR
+12V
U1602E
V+
V-
OPA4197IPWR
-12V
U1603A
C5c0054_T1
OSC_OCT1
OSC_OCT1
OPA4197IPWR
U1603B
C5c0054_T2
OSC_OCT2
OSC_OCT2
OPA4197IPWR
U1603C
C5c0054_T3
OSC_OCT3
OSC_OCT3
OPA4197IPWR
U1603D
C5c0054_T4
OSC_OCT4
OSC_OCT4
OPA4197IPWR
+12V
U1603E
V+
V-
OPA4197IPWR
-12V
U1604A
OCT_TOP
OSC_OCT5
OSC_OCT5
OPA4197IPWR
U1604B
C5c0054_ONE_V
C5c0054_GAIN_SUM
OCT_TOP
OPA4197IPWR
U1604C
C5c0054_ONE_RAW
C5c0054_ONE_V
C5c0054_ONE_V
OPA4197IPWR
U1604D
AGND
C5c0054_N2_SUM
OCT_BOTTOM
OPA4197IPWR
+12V
U1604E
V+
V-
OPA4197IPWR
-12V
C1603
+12V
AGND
100 nF
C1604
+12V
AGND
100 nF
C1605
-12V
AGND
100 nF
C1606
+12V
AGND
100 nF
C1607
-12V
AGND
100 nF
C1608
+12V
AGND
100 nF
C1609
-12V
AGND
100 nF

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Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.6
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Sheet: /F1/
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Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 10/36

Sheet: /F2/
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Size: A0 Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 11/36

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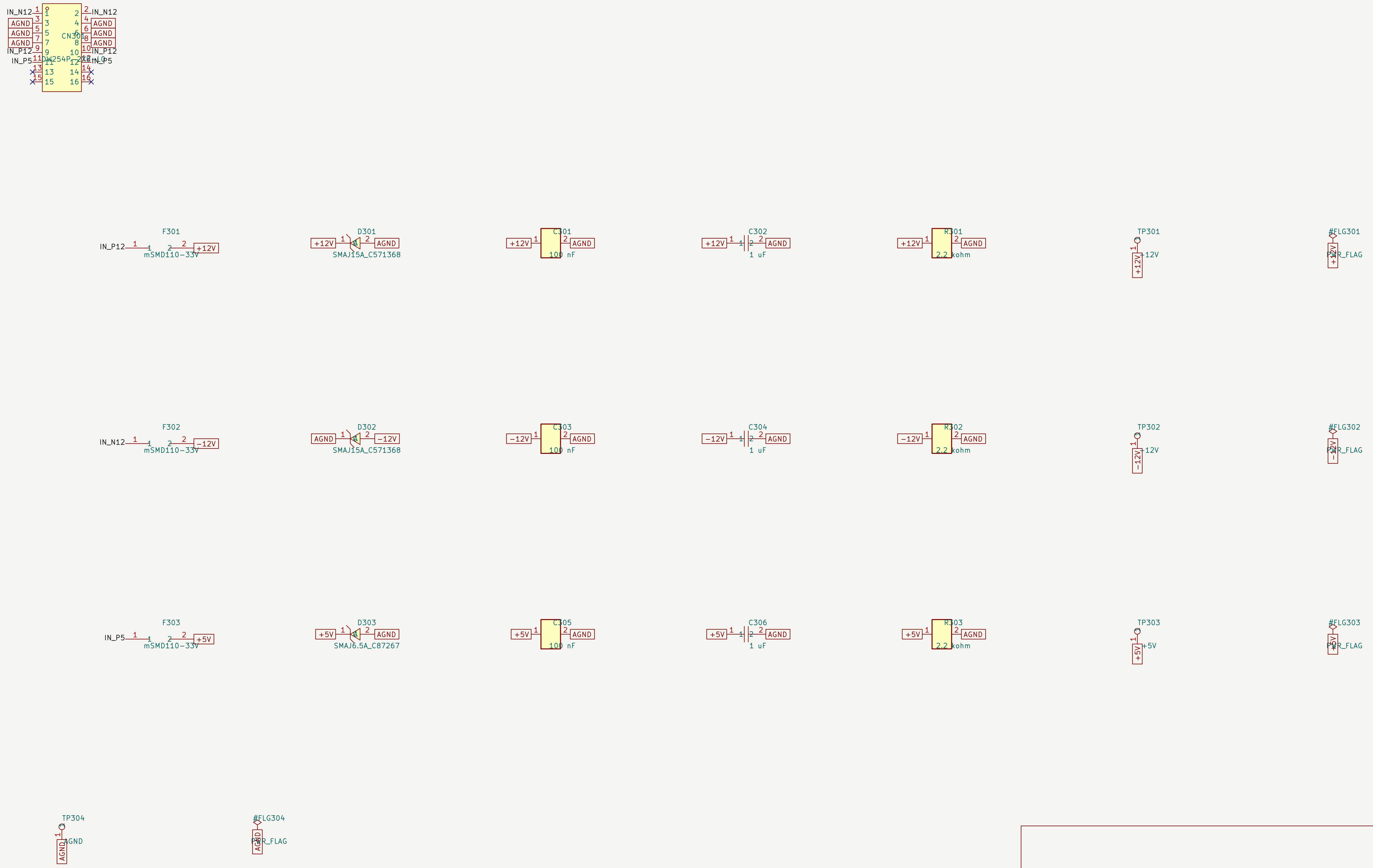
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 12/36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AGND
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IN_P12
IN_P5
IN_P5
F301
IN_P12
+12V
mSMD110-33V
D301
+12V
AGND
SMAJ15A_C571368
C301
+12V
AGND
100 nF
C302
+12V
AGND
1 uF
R301
+12V
AGND
2.2 kohm
TP301
+12V
+12V
#FLG301
+12V
PWR_FLAG
F302
IN_N12
-12V
mSMD110-33V
D302
AGND
-12V
SMAJ15A_C571368
C303
-12V
AGND
100 nF
C304
-12V
AGND
1 uF
R302
-12V
AGND
2.2 kohm
TP302
-12V
-12V
#FLG302
-12V
PWR_FLAG
F303
IN_P5
+5V
mSMD110-33V
D303
+5V
AGND
SMAJ6.5A_C87267
C305
+5V
AGND
100 nF
C306
+5V
AGND
1 uF
R303
+5V
AGND
2.2 kohm
TP303
+5V
+5V
#FLG303
+5V
PWR_FLAG
TP304
AGND
AGND
#FLG304
AGND
PWR_FLAG

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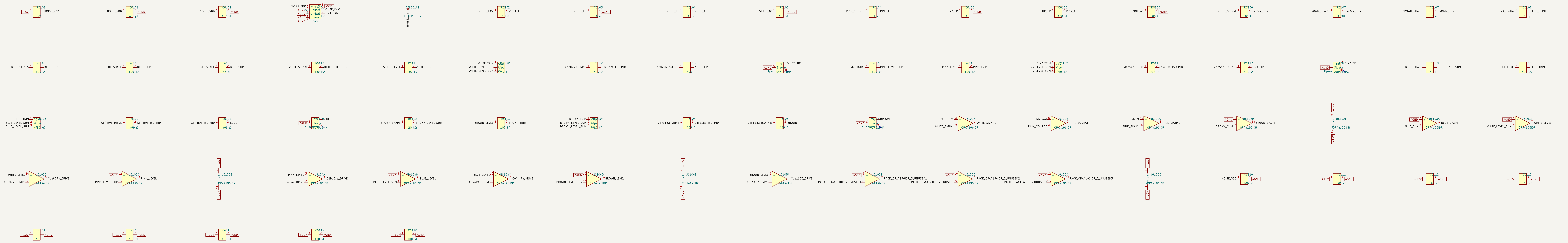
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 13/36


R6101 +5V NOISE_VDD 22 Ω
C6101 NOISE_VDD AGND 4.7 µF
C6102 NOISE_VDD AGND 100 nF
U6101 NOISE2 NOISE_VDD +5V WHITE_OUT PINK_OUT Unused Unused AGND WHITE_RAW PINK_RAW
#FLG6101 FILTERED_5V NOISE_VDD
R6102 WHITE_RAW WHITE_LP 1 kΩ
C6103 WHITE_LP AGND 10 nF
C6104 WHITE_LP WHITE_AC 100 nF
R6103 WHITE_AC AGND 100 kΩ
R6104 PINK_SOURCE PINK_LP 1 kΩ
C6105 PINK_LP AGND 10 nF
C6106 PINK_LP PINK_AC 100 nF
R6105 PINK_AC AGND 100 kΩ
R6106 WHITE_SIGNAL BROWN_SUM 100 kΩ
R6107 BROWN_SHAPE BROWN_SUM 1 MΩ
C6107 BROWN_SHAPE BROWN_SUM 10 nF
C6108 PINK_SIGNAL BLUE_SERIES 100 pF
R6108 BLUE_SERIES BLUE_SUM 100 kΩ
R6109 BLUE_SHAPE BLUE_SUM 100 kΩ
C6109 BLUE_SHAPE BLUE_SUM 33 pF
R6110 WHITE_SIGNAL WHITE_LEVEL_SUM 100 kΩ
R6111 WHITE_LEVEL WHITE_TRIM 100 kΩ
RV6101 WHITE_TRIM WHITE_LEVEL_SUM WHITE_LEVEL_SUM CCW Wiper CW 10 kΩ
R6112 Cbe877b_DRIVE Cbe877b_ISO_MID 499 Ω
R6113 Cbe877b_ISO_MID WHITE_TIP 499 Ω
J6109 AGND Tip Sleeve T S TN WHITE_TIP Tip−normal WQP518MA
R6114 PINK_SIGNAL PINK_LEVEL_SUM 100 kΩ
R6115 PINK_LEVEL PINK_TRIM 100 kΩ
RV6102 PINK_TRIM PINK_LEVEL_SUM PINK_LEVEL_SUM CCW Wiper CW 10 kΩ
R6116 Cdbc5aa_DRIVE Cdbc5aa_ISO_MID 499 Ω
R6117 Cdbc5aa_ISO_MID PINK_TIP 499 Ω
J6209 AGND Tip Sleeve PINK_TIP Tip−normal WQP518MA
R6118 BLUE_SHAPE BLUE_LEVEL_SUM 10 kΩ
R6119 BLUE_LEVEL BLUE_TRIM 100 kΩ
RV6103 BLUE_TRIM BLUE_LEVEL_SUM BLUE_LEVEL_SUM CCW Wiper CW 10 kΩ
R6120 Ce44f9a_DRIVE Ce44f9a_ISO_MID 499 Ω
R6121 Ce44f9a_ISO_MID BLUE_TIP 499 Ω
J6110 AGND Tip Sleeve BLUE_TIP Tip−normal WQP518MA
R6122 BROWN_SHAPE BROWN_LEVEL_SUM 20 kΩ
R6123 BROWN_LEVEL BROWN_TRIM 100 kΩ
RV6104 BROWN_TRIM BROWN_LEVEL_SUM BROWN_LEVEL_SUM CCW Wiper CW 10 kΩ
R6124 Cde1183_DRIVE Cde1183_ISO_MID 499 Ω
R6125 Cde1183_ISO_MID BROWN_TIP 499 Ω
J6210 AGND Tip Sleeve BROWN_TIP Tip−normal WQP518MA
U6102A OPA4196IDR WHITE_AC WHITE_SIGNAL WHITE_SIGNAL
U6102B OPA4196IDR PINK_RAW PINK_SOURCE PINK_SOURCE
U6102C OPA4196IDR PINK_AC PINK_SIGNAL PINK_SIGNAL
U6102D OPA4196IDR AGND BROWN_SUM BROWN_SHAPE
U6102E OPA4196IDR +12V −12V V+ V−
U6103A OPA4196IDR AGND BLUE_SUM BLUE_SHAPE
U6103B OPA4196IDR AGND WHITE_LEVEL_SUM WHITE_LEVEL
U6103C OPA4196IDR WHITE_LEVEL Cbe877b_DRIVE Cbe877b_DRIVE
U6103D OPA4196IDR AGND PINK_LEVEL_SUM PINK_LEVEL
U6103E OPA4196IDR +12V −12V
U6104A OPA4196IDR PINK_LEVEL Cdbc5aa_DRIVE Cdbc5aa_DRIVE
U6104B OPA4196IDR AGND BLUE_LEVEL_SUM BLUE_LEVEL
U6104C OPA4196IDR BLUE_LEVEL Ce44f9a_DRIVE Ce44f9a_DRIVE
U6104D OPA4196IDR AGND BROWN_LEVEL_SUM BROWN_LEVEL
U6104E OPA4196IDR +12V −12V
U6105A OPA4196IDR BROWN_LEVEL Cde1183_DRIVE Cde1183_DRIVE
U6105B OPA4196IDR AGND PACK_OPA4196IDR_3_UNUSED1 PACK_OPA4196IDR_3_UNUSED1
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U6105D OPA4196IDR AGND PACK_OPA4196IDR_3_UNUSED3 PACK_OPA4196IDR_3_UNUSED3
U6105E OPA4196IDR +12V −12V
C6110 NOISE_VDD AGND 100 nF
C6111 +12V AGND 100 nF
C6112 −12V AGND 100 nF
C6113 +12V AGND 100 nF
C6114 −12V AGND 100 nF
C6115 +12V AGND 100 nF
C6116 −12V AGND 100 nF
C6117 +12V AGND 100 nF
C6118 −12V AGND 100 nF

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Size: A0 Date:
KiCad E.D.A. 10.0.6
Rev:
Id: 14/36

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Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 15/36

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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 16/36

Sheet: /E3/
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Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 17/36

Sheet: /E4/
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Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 18/36

Sheet: /E5/
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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 19/36

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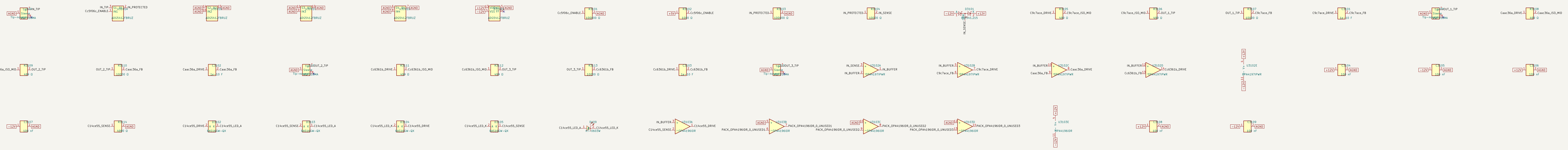
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 20/36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Sleeve
Tip-normal
WQP518MA
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IN_TIP
Cc5f06c_ENABLE
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U3101B
ADG5412FBRUZ
U3101C
ADG5412FBRUZ
U3101D
ADG5412FBRUZ
+12V
-12V
VDD
VSS
GND
U3101E
ADG5412FBRUZ
R3101
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R3102
+5V
1000 Ω
R3103
100000 Ω
R3104
10000 Ω
IN_SENSE
D3101
BAT54S,215
R3105
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3106
OUT_1_TIP
499 Ω
R3107
C9c7ace_FB
10000 Ω
C3101
1e-10 F
R3108
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3109
OUT_2_TIP
499 Ω
R3110
Caac36a_FB
10000 Ω
C3102
1e-10 F
R3111
Cc6361b_DRIVE
Cc6361b_ISO_MID
499 Ω
R3112
OUT_3_TIP
499 Ω
R3113
Cc6361b_FB
10000 Ω
C3103
1e-10 F
U3102A
IN_BUFFER
OPA4197IPWR
U3102B
U3102C
U3102D
U3102E
OPA4197IPWR
C3104
100 nF
C3105
100 nF
C3106
100 nF
C3107
100 nF
R3114
C14ce55_SENSE
4990 Ω
D3102
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3103
BAS16GW-QX
D3104
C14ce55_LED_K
BAS16GW-QX
D3105
BAS16GW-QX
D109
KT-0603W
U3103A
OPA4196IDR
U3103B
PACK_OPA4196IDR_0_UNUSED1
U3103C
PACK_OPA4196IDR_0_UNUSED2
U3103D
PACK_OPA4196IDR_0_UNUSED3
U3103E
OPA4196IDR
C3108
100 nF
C3109
100 nF

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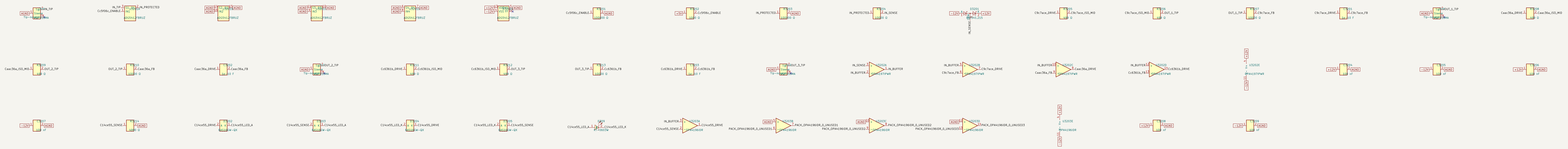
AGND
Tip
Sleeve
IN_TIP
IN_TIP
Cc5f06c_ENABLE
IN_PROTECTED
U3201A
ADG5412FBRUZ
U3201B
ADG5412FBRUZ
U3201C
ADG5412FBRUZ
U3201D
ADG5412FBRUZ
+12V
-12V
VDD
VSS
U3201E
ADG5412FBRUZ
R3201
100000 Ω
R3202
+5V
1000 Ω
R3203
100000 Ω
R3204
IN_SENSE
10000 Ω
D3201
BAT54S,215
R3205
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3206
OUT_1_TIP
499 Ω
R3207
C9c7ace_FB
10000 Ω
C3201
1e-10 F
R3208
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3209
OUT_2_TIP
499 Ω
R3210
Caac36a_FB
10000 Ω
C3202
1e-10 F
R3211
Cc6361b_DRIVE
Cc6361b_ISO_MID
499 Ω
R3212
OUT_3_TIP
499 Ω
R3213
Cc6361b_FB
10000 Ω
C3203
1e-10 F
U3202A
IN_BUFFER
OPA4197IPWR
U3202B
U3202C
U3202D
U3202E
C3204
100 nF
C3205
100 nF
C3206
100 nF
C3207
100 nF
R3214
C14ce55_SENSE
4990 Ω
D3202
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3203
BAS16GW-QX
D3204
C14ce55_LED_K
BAS16GW-QX
D3205
BAS16GW-QX
KT-0603W
U3203A
OPA4196IDR
U3203B
PACK_OPA4196IDR_0_UNUSED1
U3203C
PACK_OPA4196IDR_0_UNUSED2
U3203D
PACK_OPA4196IDR_0_UNUSED3
U3203E
C3208
100 nF
C3209
100 nF

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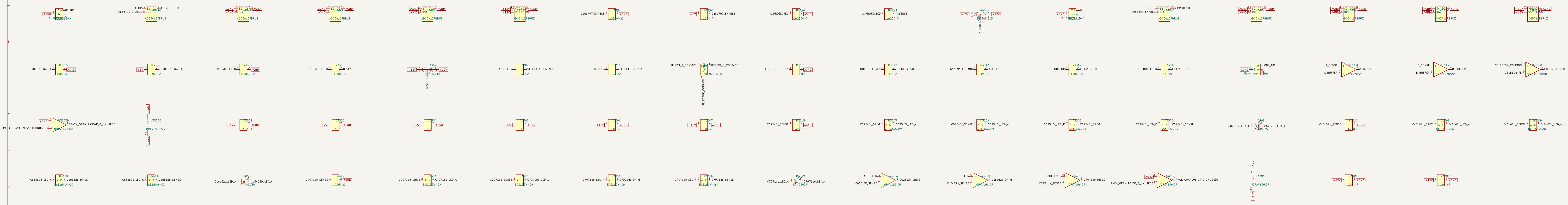
AGND
A_TIP
Tip
Sleeve
Tip-normal
A_TIP
Cadd797_ENABLE
A_PROTECTED
S1 D1
IN1
U3301A
ADG5412FBRUZ
S2 D2
IN2
U3301B
ADG5412FBRUZ
S3 D3
IN3
U3301C
ADG5412FBRUZ
S4 D4
IN4
U3301D
ADG5412FBRUZ
+12V
-12V
VDD GND
VSS FF
U3301E
ADG5412FBRUZ
Cadd797_ENABLE
R3301
100000 Ω
+5V
R3302
Cadd797_ENABLE
1000 Ω
A_PROTECTED
R3303
100000 Ω
A_PROTECTED
R3304
A_SENSE
10000 Ω
D3301
BAT54S,215
A_SENSE
B_TIP
B_TIP
C9a8343_ENABLE
B_PROTECTED
U3302A
U3302B
U3302C
U3302D
U3302E
ADG5412FBRUZ
C9a8343_ENABLE
R3305
100000 Ω
+5V
R3306
C9a8343_ENABLE
1000 Ω
B_PROTECTED
R3307
100000 Ω
B_PROTECTED
R3308
B_SENSE
10000 Ω
D3302
BAT54S,215
B_SENSE
A_BUFFER
R3309
SELECT_A_CONTACT
2.2 kΩ
B_BUFFER
R3310
SELECT_B_CONTACT
2.2 kΩ
SELECT_A_CONTACT
SELECT_B_CONTACT
2MS1T2B1M2QES-5
SELECTOR_COMMON
SELECTOR_COMMON
R3311
10 MΩ
OUT_BUFFERED
R3312
C61b194_ISO_MID
499 Ω
C61b194_ISO_MID
R3313
OUT_TIP
499 Ω
OUT_TIP
R3314
C61b194_FB
10000 Ω
OUT_BUFFERED
C3301
C61b194_FB
1e-10 F
OUT_TIP
A_SENSE
A_BUFFER
U3303A
OPA4197IPWR
A_BUFFER
B_SENSE
B_BUFFER
U3303B
OPA4197IPWR
B_BUFFER
SELECTOR_COMMON
C61b194_FB
U3303C
OPA4197IPWR
OUT_BUFFERED
PACK_OPA4197IPWR_0_UNUSED3
U3303D
OPA4197IPWR
PACK_OPA4197IPWR_0_UNUSED3
U3303E
V+ V-
OPA4197IPWR
C3302
100 nF
C3303
100 nF
C3304
100 nF
C3305
100 nF
C3306
100 nF
C3307
100 nF
C025c3f_SENSE
R3315
4990 Ω
C025c3f_DRIVE
D3303
C025c3f_LED_A
BAS16GW-QX
C025c3f_SENSE
D3304
C025c3f_LED_A
BAS16GW-QX
C025c3f_LED_K
D3305
C025c3f_DRIVE
BAS16GW-QX
C025c3f_LED_K
D3306
C025c3f_SENSE
BAS16GW-QX
C025c3f_LED_A
D909
C025c3f_LED_K
KT-0603W
Cc6c62b_SENSE
R3316
4990 Ω
Cc6c62b_DRIVE
D3308
Cc6c62b_LED_A
BAS16GW-QX
Cc6c62b_SENSE
D3309
Cc6c62b_LED_A
BAS16GW-QX
Cc6c62b_LED_K
D3310
Cc6c62b_DRIVE
BAS16GW-QX
Cc6c62b_LED_K
D3311
Cc6c62b_SENSE
BAS16GW-QX
Cc6c62b_LED_A
D909
Cc6c62b_LED_K
KT-0603W
C7971de_SENSE
R3317
4990 Ω
C7971de_DRIVE
D3313
C7971de_LED_A
BAS16GW-QX
C7971de_SENSE
D3314
C7971de_LED_A
BAS16GW-QX
C7971de_LED_K
D3315
C7971de_DRIVE
BAS16GW-QX
C7971de_LED_K
D3316
C7971de_SENSE
BAS16GW-QX
C7971de_LED_A
D1009
C7971de_LED_K
KT-0603W
A_BUFFER
C025c3f_SENSE
U3304A
OPA4196IDR
C025c3f_DRIVE
B_BUFFER
Cc6c62b_SENSE
U3304B
OPA4196IDR
Cc6c62b_DRIVE
OUT_BUFFERED
C7971de_SENSE
U3304C
OPA4196IDR
C7971de_DRIVE
PACK_OPA4196IDR_0_UNUSED3
U3304D
OPA4196IDR
PACK_OPA4196IDR_0_UNUSED3
U3304E
OPA4196IDR
C3308
100 nF
C3309
100 nF

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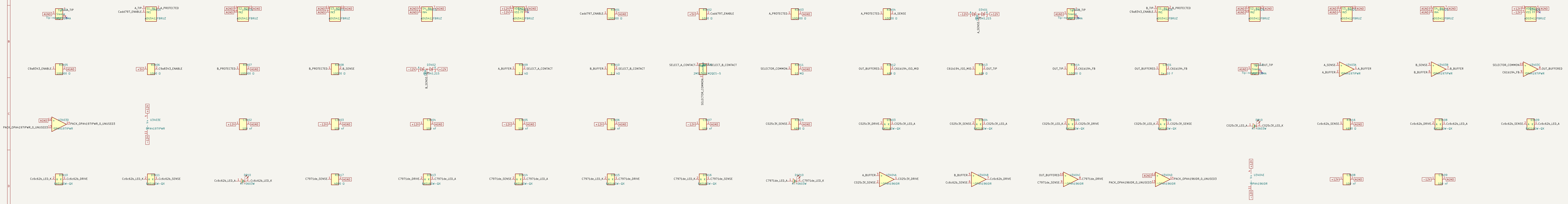
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Tip
Sleeve
Tip-normal
WQP518MA
AGND
U3401A
A_TIP
Cadd797_ENABLE
A_PROTECTED
ADG5412FBRUZ
U3401B
U3401C
U3401D
U3401E
VDD
VSS
GND
+12V
-12V
R3401
Cadd797_ENABLE
100000 Ω
R3402
+5V
1000 Ω
R3403
A_PROTECTED
R3404
A_SENSE
10000 Ω
D3401
BAV99S,215
J910
B_TIP
U3402A
C9a8343_ENABLE
B_PROTECTED
U3402B
U3402C
U3402D
U3402E
R3405
R3406
R3407
B_PROTECTED
R3408
B_SENSE
D3402
R3409
A_BUFFER
SELECT_A_CONTACT
2.2 kΩ
R3410
B_BUFFER
SELECT_B_CONTACT
SW?
2MS1T1B1M2QES-5
SELECTOR_COMMON
R3411
10 MΩ
R3412
OUT_BUFFERED
C61b194_ISO_MID
499 Ω
R3413
OUT_TIP
R3414
C61b194_FB
C3401
1e-10 F
J1010
U3403A
A_SENSE
OPA4197IPWR
U3403B
U3403C
U3403D
PACK_OPA4197IPWR_0_UNUSED3
U3403E
V+
V-
C3402
100 nF
C3403
C3404
C3405
C3406
C3407
R3415
C025c3f_SENSE
4990 Ω
D3403
C025c3f_DRIVE
C025c3f_LED_A
BAS16GW-QX
D3404
D3405
C025c3f_LED_K
D3406
D810
KT-0603W
R3416
Cc6c62b_SENSE
D3408
Cc6c62b_DRIVE
Cc6c62b_LED_A
D3409
D3410
Cc6c62b_LED_K
D3411
D910
R3417
C7971de_SENSE
D3413
C7971de_DRIVE
C7971de_LED_A
D3414
D3415
C7971de_LED_K
D3416
D1010
U3404A
OPA4196IDR
U3404B
U3404C
U3404D
PACK_OPA4196IDR_0_UNUSED3
U3404E
C3408
C3409

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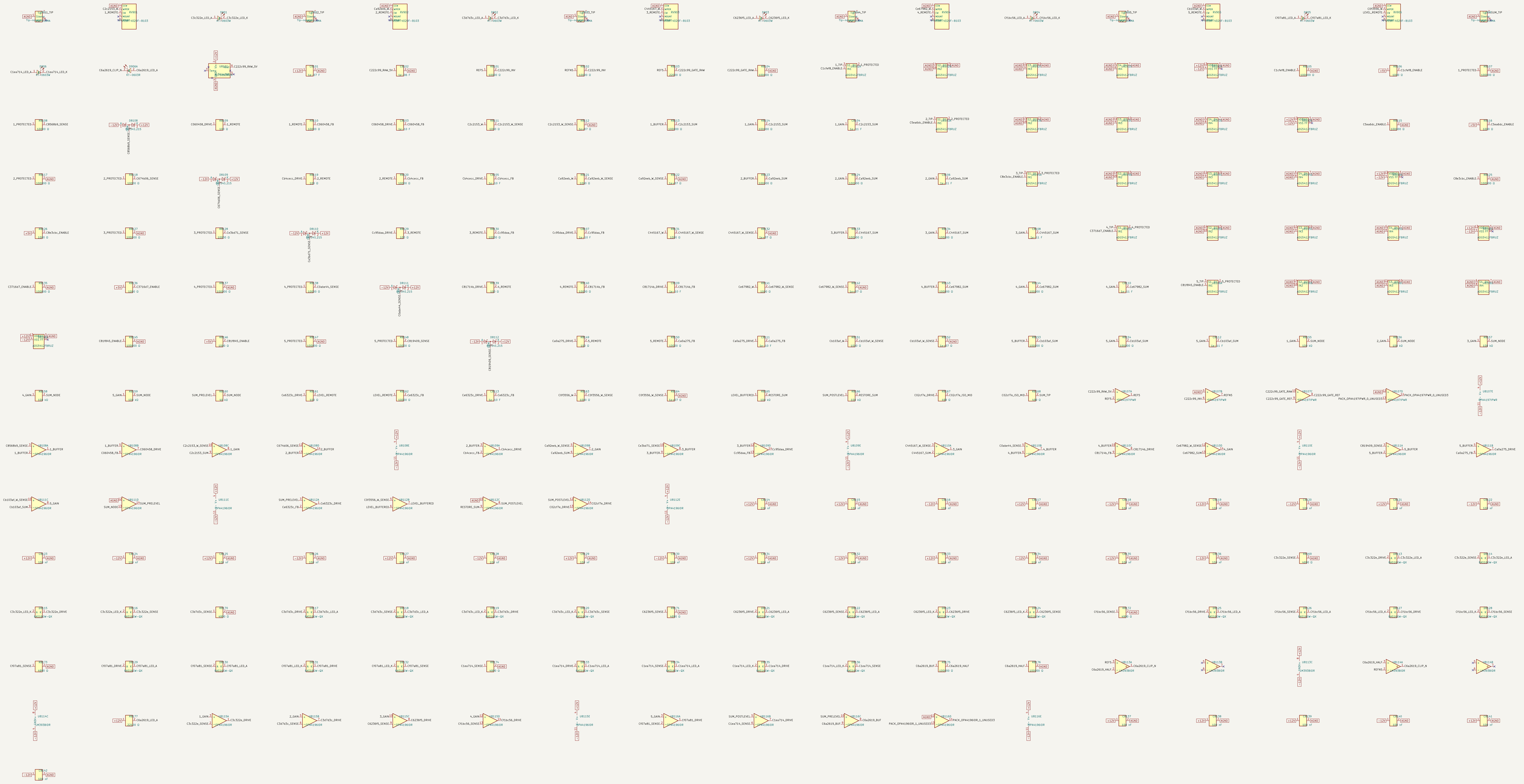

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Size: A0
Date:
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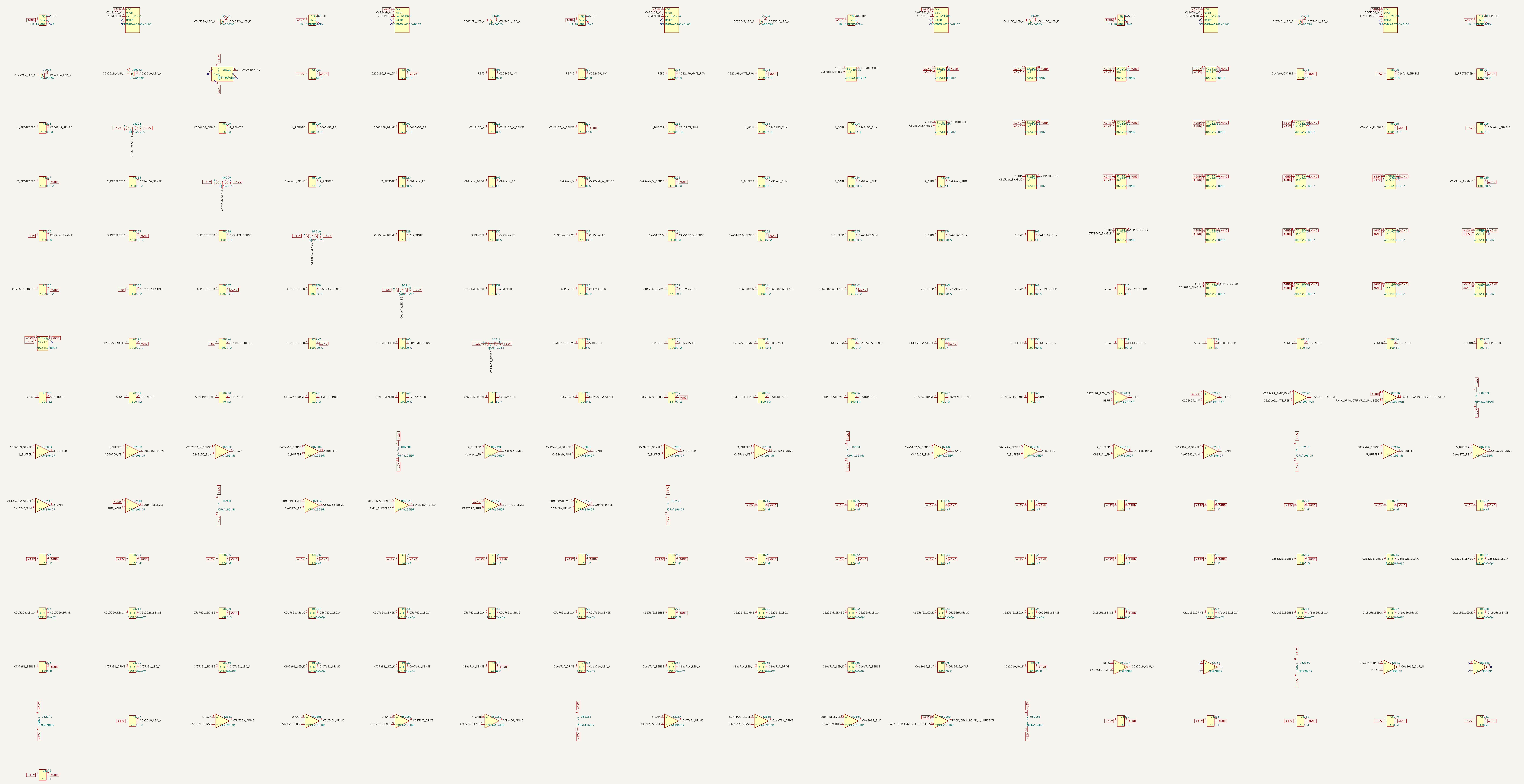

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Size: A0 Date: **Rev:**
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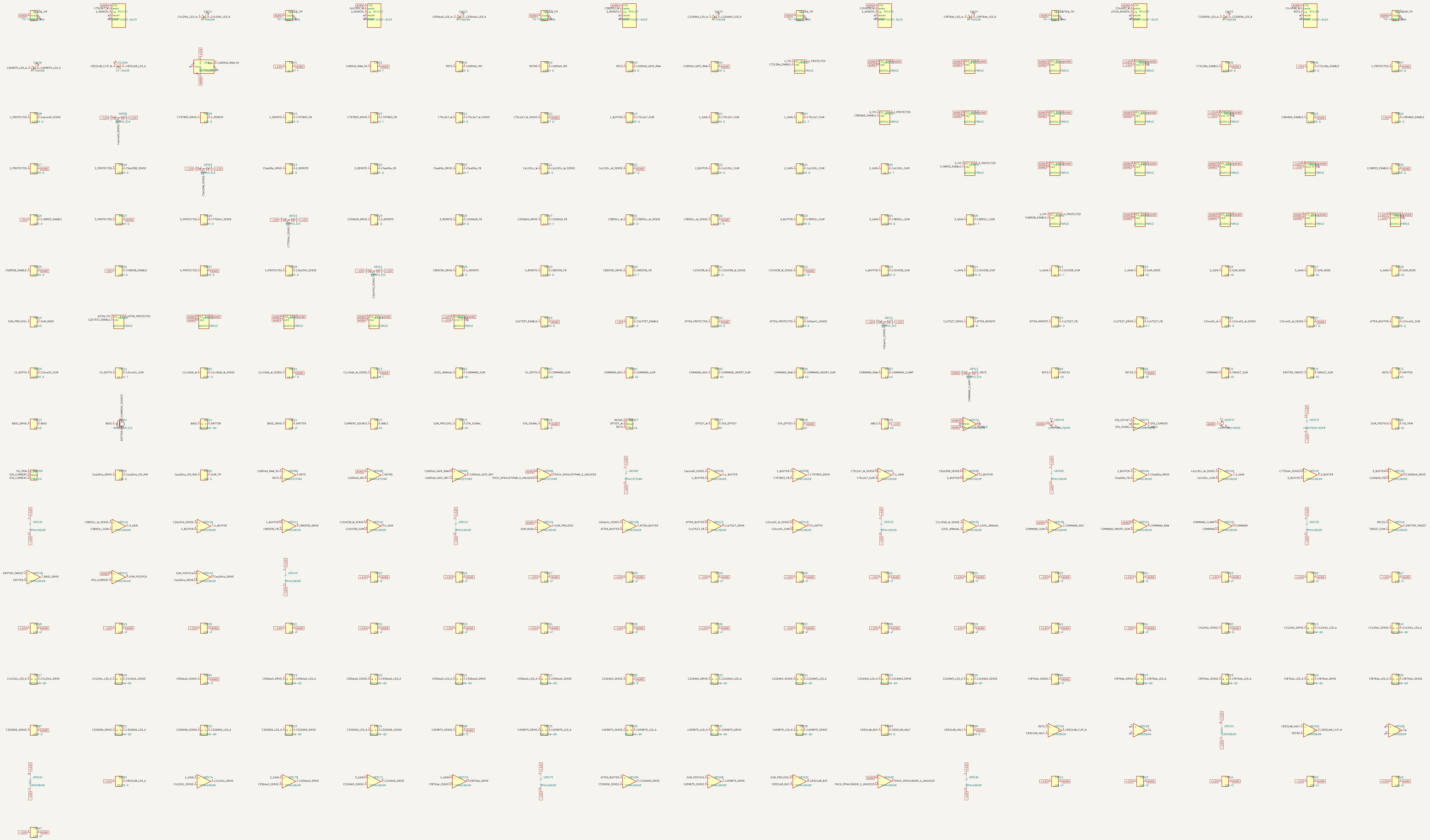

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Date:
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KiCad E.D.A. 10.0.6
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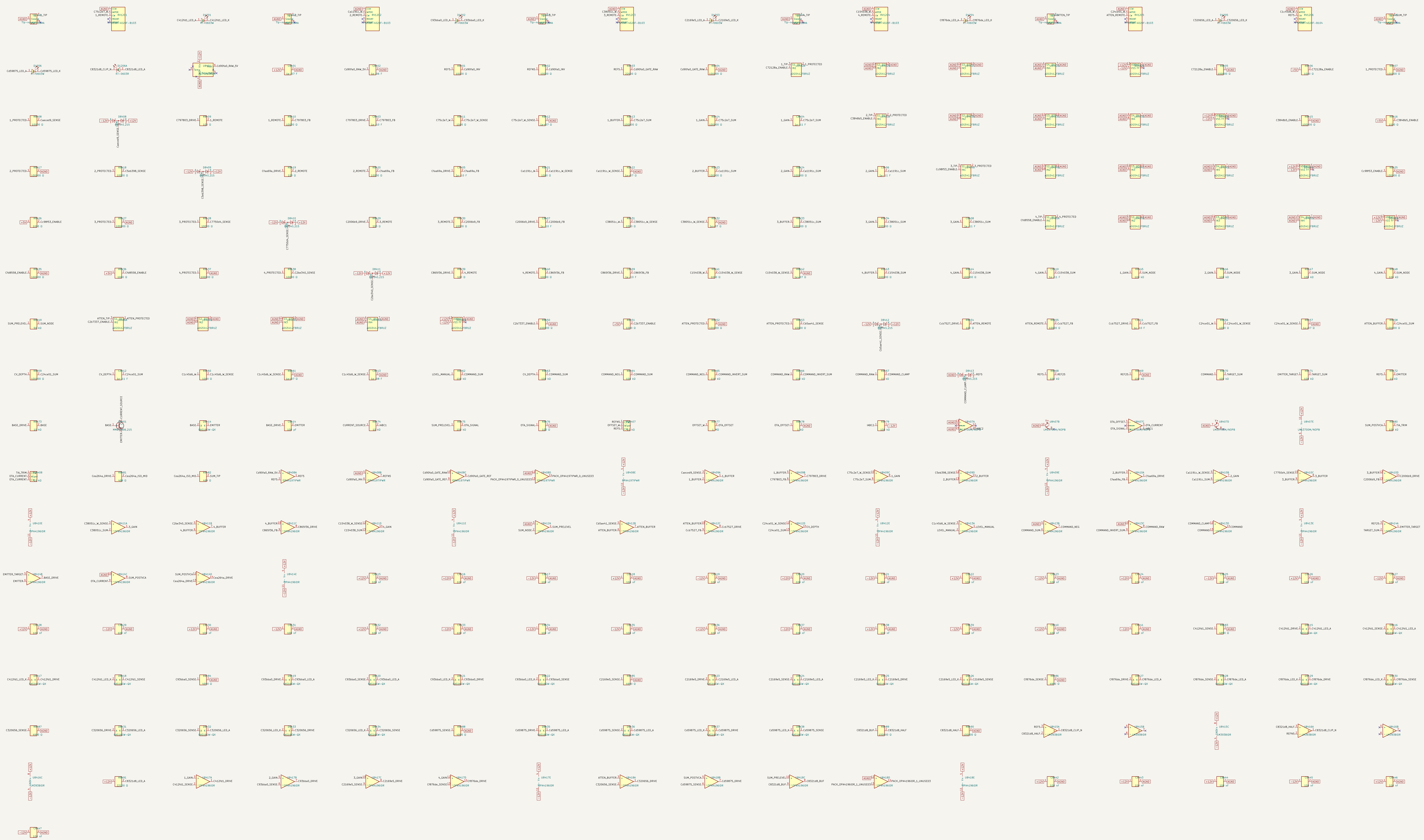

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Size: A0	Date:	**Rev:**
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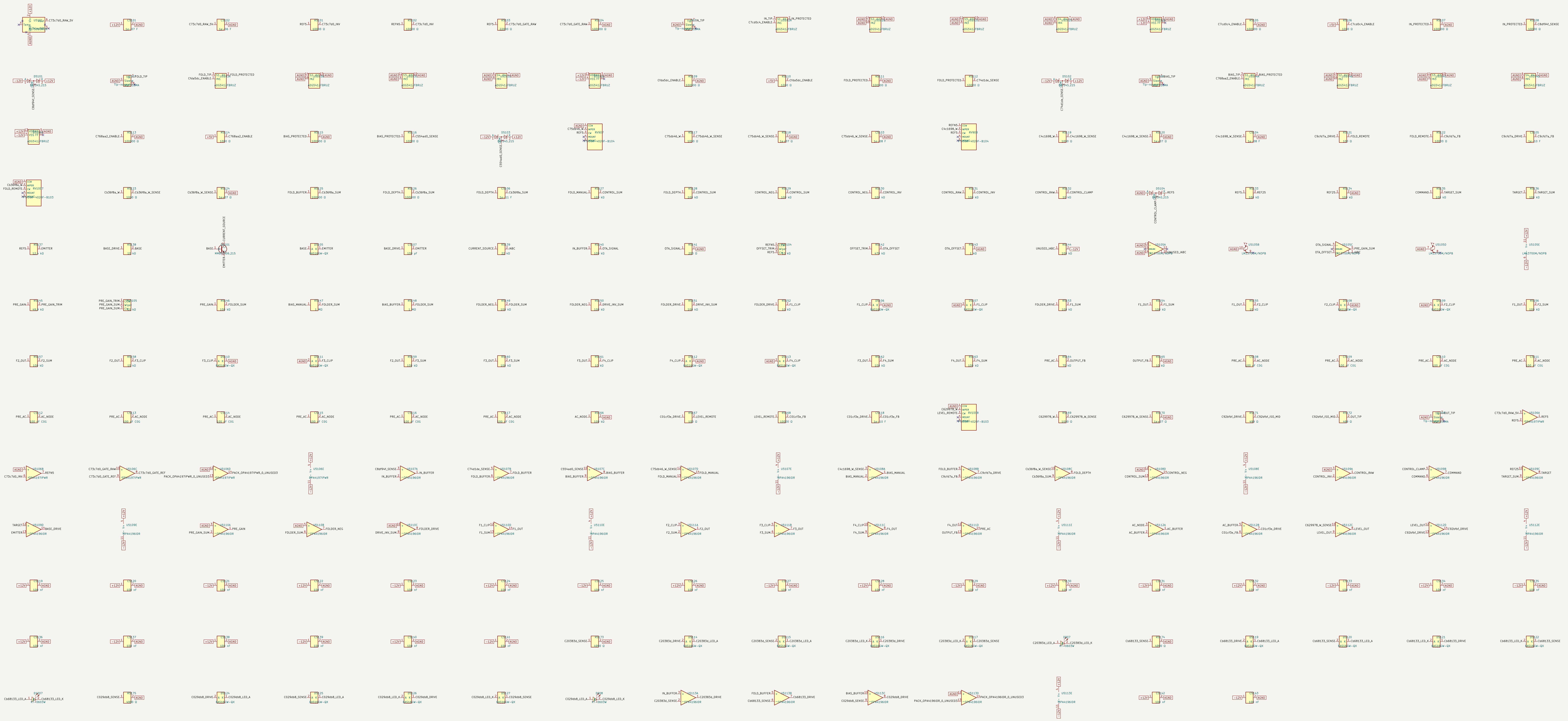

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Size: A0 | Date: | **Rev:**
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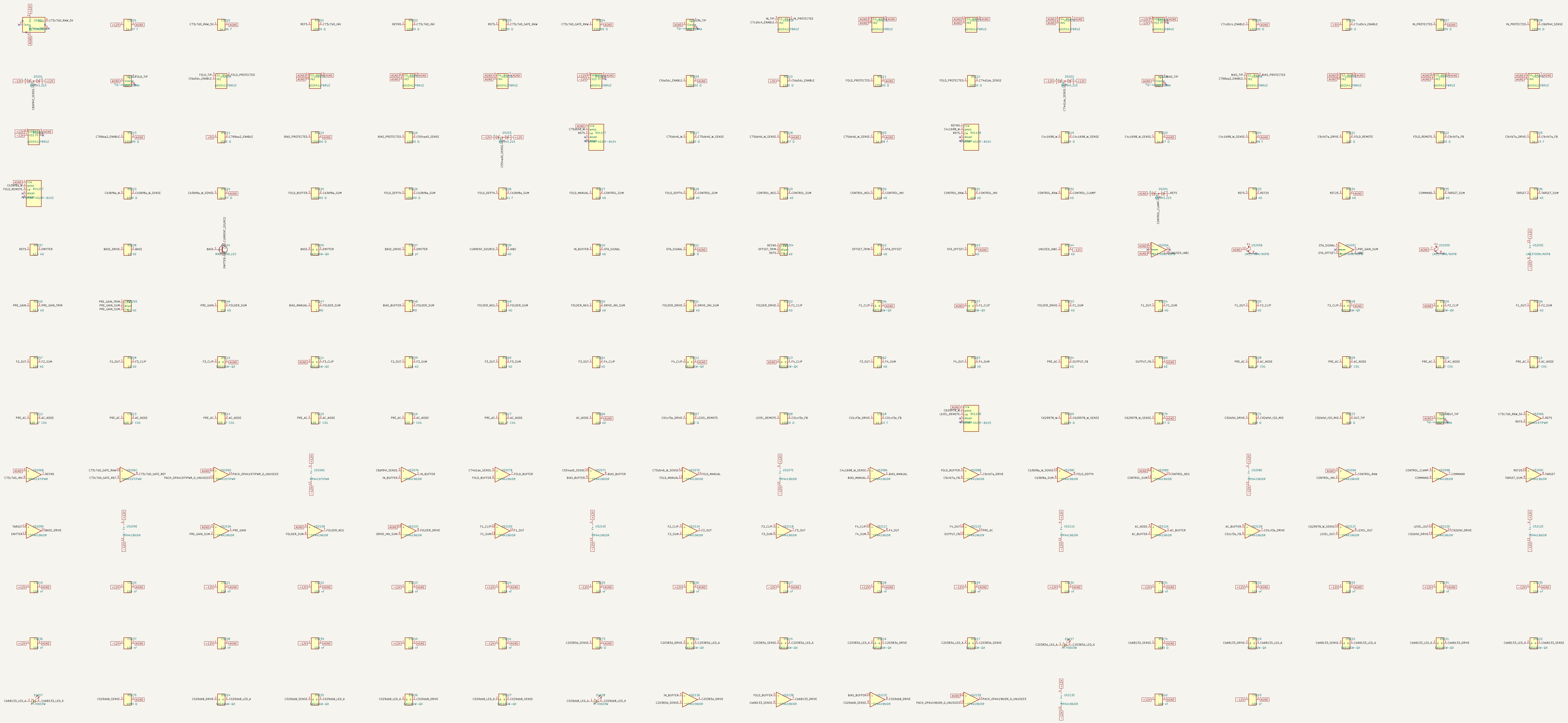

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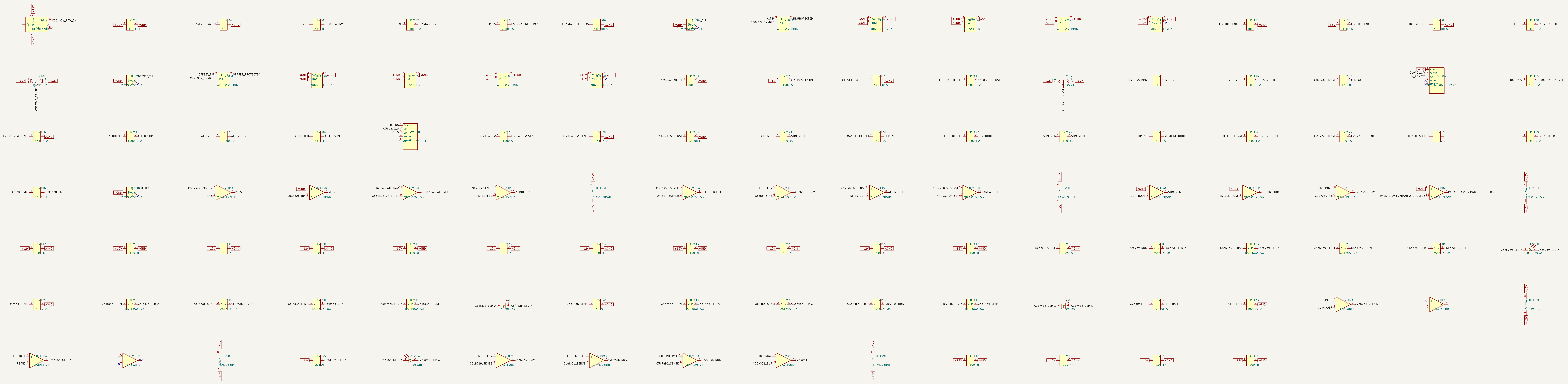
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 30/36



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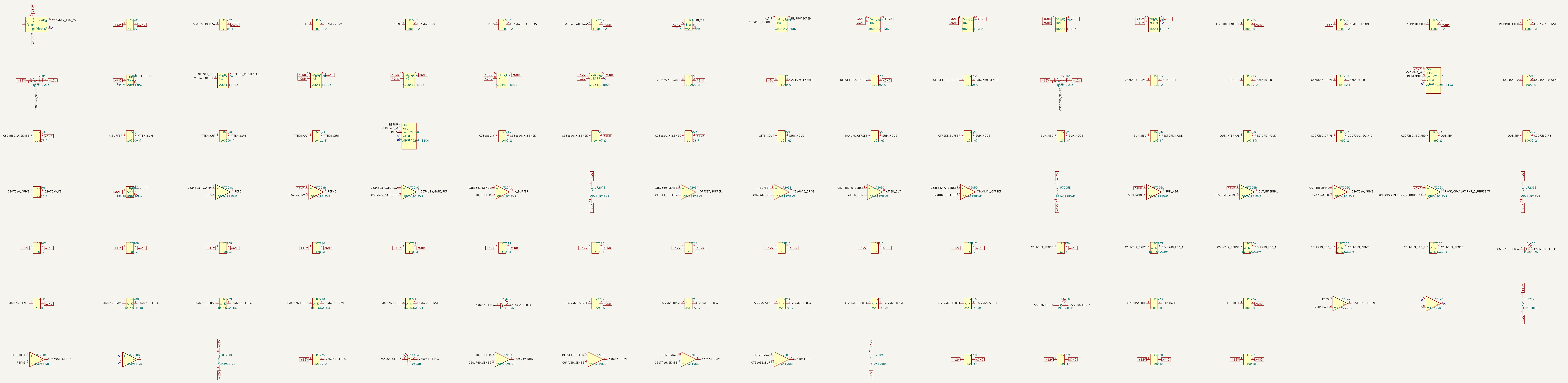
KiCad E.D.A. 10.0.6 | Id: 31/36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C7201 1e−07 F +12V AGND
C7202 1e−06 F C534b2a_RAW_5V AGND
R7201 10000 Ω REF5 C534b2a_INV
R7202 10000 Ω REFN5 C534b2a_INV
R7203 22000 Ω REF5 C534b2a_GATE_RAW
R7204 100000 Ω C534b2a_GATE_RAW AGND
Tip Sleeve Tip-normal AGND IN_TIP
U7202A ADG5412FBRUZ S1 D1 IN1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7202B ADG5412FBRUZ S2 D2 IN2 AGND
U7202C ADG5412FBRUZ S3 D3 IN3 AGND
U7202D ADG5412FBRUZ S4 D4 IN4 AGND
U7202E ADG5412FBRUZ VDD GND VSS FF +12V −12V AGND
R7205 100000 Ω C38d00f_ENABLE AGND
R7206 1000 Ω +5V C38d00f_ENABLE
R7207 100000 Ω IN_PROTECTED AGND
R7208 10000 Ω IN_PROTECTED C3833e3_SENSE
D7201 BAT54S,215 A K COM −12V +12V C3833e3_SENSE
Tip Sleeve Tip-normal AGND OFFSET_TIP
U7203A ADG5412FBRUZ S1 D1 IN1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7203B ADG5412FBRUZ S2 D2 IN2 AGND
U7203C ADG5412FBRUZ S3 D3 IN3 AGND
U7203D ADG5412FBRUZ S4 D4 IN4 AGND
U7203E ADG5412FBRUZ VDD GND VSS FF +12V −12V AGND
R7209 100000 Ω C27197a_ENABLE AGND
R7210 1000 Ω +5V C27197a_ENABLE
R7211 100000 Ω OFFSET_PROTECTED AGND
R7212 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7202 BAT54S,215 A K COM −12V +12V C360350_SENSE
R7213 100 Ω C8e6645_DRIVE IN_REMOTE
R7214 10000 Ω IN_REMOTE C8e6645_FB
C7203 1e−10 F C8e6645_DRIVE C8e6645_FB
RV1407 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7215 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7216 1e+07 Ω Cc045d2_W_SENSE AGND
R7217 100000 Ω IN_BUFFER ATTEN_SUM
R7218 100000 Ω ATTEN_OUT ATTEN_SUM
C7204 1e−11 F ATTEN_OUT ATTEN_SUM
RV1408 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7219 1000 Ω C38cac0_W C38cac0_W_SENSE
R7220 1e+07 Ω C38cac0_W_SENSE AGND
C7205 1e−08 F C38cac0_W_SENSE AGND
R7221 100 kΩ ATTEN_OUT SUM_NODE
R7222 100 kΩ MANUAL_OFFSET SUM_NODE
R7223 100 kΩ OFFSET_BUFFER SUM_NODE
R7224 100 kΩ SUM_NEG SUM_NODE
R7225 100 kΩ SUM_NEG RESTORE_NODE
R7226 100 kΩ OUT_INTERNAL RESTORE_NODE
R7227 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7228 499 Ω C2073e0_ISO_MID OUT_TIP
R7229 10000 Ω OUT_TIP C2073e0_FB
C7206 1e−10 F C2073e0_DRIVE C2073e0_FB
Tip Sleeve Tip-normal AGND OUT_TIP
U7204A OPA4197IPWR C534b2a_RAW_5V REF5
U7204B OPA4197IPWR AGND C534b2a_INV REFN5
U7204C OPA4197IPWR C534b2a_GATE_RAW C534b2a_GATE_REF
U7204D OPA4197IPWR C3833e3_SENSE IN_BUFFER
U7204E OPA4197IPWR V+ V− +12V −12V
U7205A OPA4197IPWR C360350_SENSE OFFSET_BUFFER
U7205B OPA4197IPWR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7205C OPA4197IPWR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7205D OPA4197IPWR C38cac0_W_SENSE MANUAL_OFFSET
U7205E OPA4197IPWR V+ V− +12V −12V
U7206A OPA4197IPWR AGND SUM_NODE SUM_NEG
U7206B OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7206C OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7206D OPA4197IPWR AGND PACK_OPA4197IPWR_2_UNUSED3
U7206E OPA4197IPWR V+ V− +12V −12V
C7207 100 nF +12V AGND
C7208 100 nF +12V AGND
C7209 100 nF −12V AGND
C7210 100 nF +12V AGND
C7211 100 nF −12V AGND
C7212 100 nF +12V AGND
C7213 100 nF −12V AGND
C7214 100 nF +12V AGND
C7215 100 nF −12V AGND
C7216 100 nF +12V AGND
C7217 100 nF −12V AGND
R7230 4990 Ω C6cb7d9_SENSE AGND
D7203 BAS16GW-QX A K C6cb7d9_DRIVE C6cb7d9_LED_A
D7204 BAS16GW-QX C6cb7d9_SENSE C6cb7d9_LED_A
D7205 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_DRIVE
D7206 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_SENSE
D1408 KT-0603W C6cb7d9_LED_A C6cb7d9_LED_K
R7231 4990 Ω Cd4fa3b_SENSE AGND
D7208 BAS16GW-QX Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7209 BAS16GW-QX Cd4fa3b_SENSE Cd4fa3b_LED_A
D7210 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7211 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_SENSE
D1409 KT-0603W Cd4fa3b_LED_A Cd4fa3b_LED_K
R7232 4990 Ω C3c74b6_SENSE AGND
D7213 BAS16GW-QX C3c74b6_DRIVE C3c74b6_LED_A
D7214 BAS16GW-QX C3c74b6_SENSE C3c74b6_LED_A
D7215 BAS16GW-QX C3c74b6_LED_K C3c74b6_DRIVE
D7216 BAS16GW-QX C3c74b6_LED_K C3c74b6_SENSE
D1410 KT-0603W C3c74b6_LED_A C3c74b6_LED_K
R7233 100000 Ω C75b051_BUF CLIP_HALF
R7234 100000 Ω CLIP_HALF AGND
U7207A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7207B LM393BIDR
U7207C LM393BIDR GND V+ +12V −12V
U7208A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7208B LM393BIDR
U7208C LM393BIDR GND V+ +12V −12V
R7235 22100 Ω +12V C75b051_LED_A
D1410A KT-0603R C75b051_CLIP_N C75b051_LED_A
U7209A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7209B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7209C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7209D OPA4196IDR OUT_INTERNAL C75b051_BUF
U7209E OPA4196IDR V+ V− +12V −12V
C7218 100 nF +12V AGND
C7219 100 nF +12V AGND
C7220 100 nF +12V AGND
C7221 100 nF −12V AGND

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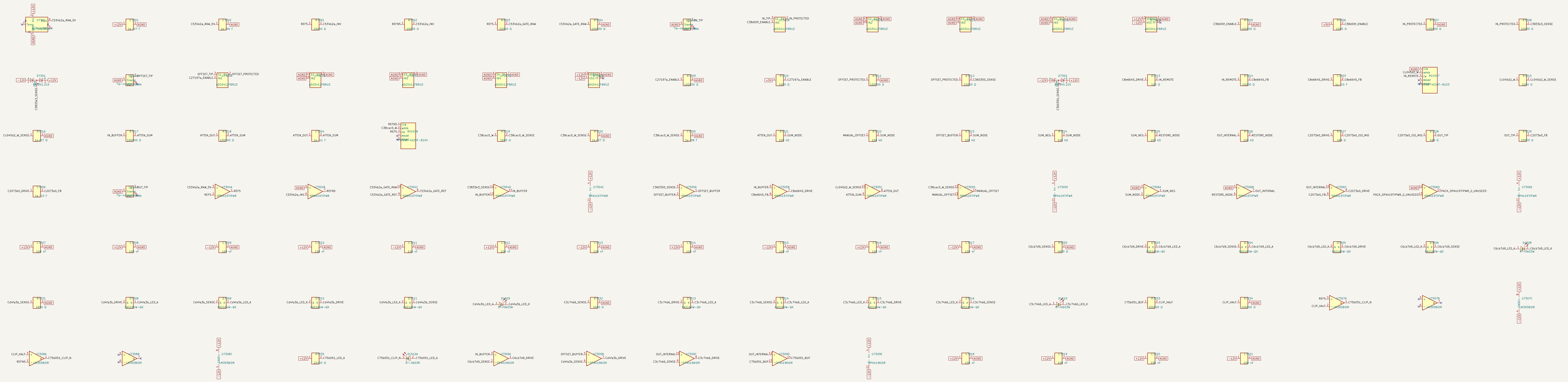
KiCad E.D.A. 10.0.6 | Id: 32/36



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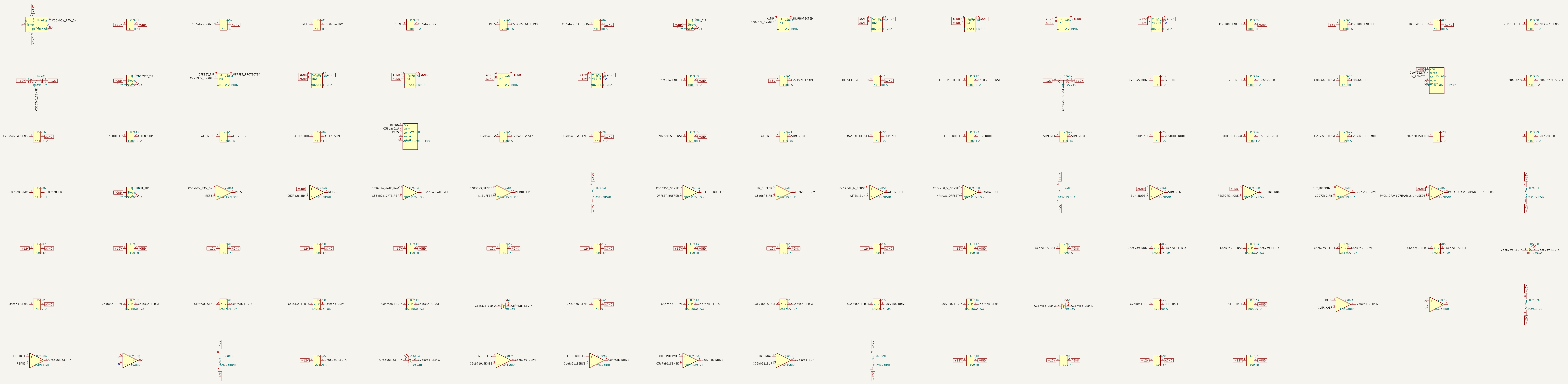
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C7401 1e-07 F +12V AGND
C7402 1e-06 F C534b2a_RAW_5V AGND
R7401 10000 Ω REF5 C534b2a_INV
R7402 10000 Ω REFN5 C534b2a_INV
R7403 22000 Ω REF5 C534b2a_GATE_RAW
R7404 100000 Ω C534b2a_GATE_RAW AGND
J7408 Tip Sleeve Tip-normal WQP518MA AGND IN_TIP
U7402A ADG5412FBRUZ S1 D1 IN1 IN_TIP IN_PROTECTED C38d00f_ENABLE
U7402B ADG5412FBRUZ S2 D2 IN2 AGND
U7402C ADG5412FBRUZ S3 D3 IN3 AGND
U7402D ADG5412FBRUZ S4 D4 IN4 AGND
U7402E ADG5412FBRUZ VDD GND VSS FF +12V −12V AGND
R7405 100000 Ω C38d00f_ENABLE AGND
R7406 1000 Ω +5V C38d00f_ENABLE
R7407 100000 Ω IN_PROTECTED AGND
R7408 10000 Ω IN_PROTECTED C3833e3_SENSE
D7401 BAT54S,215 A K COM −12V +12V C3833e3_SENSE
J7409 Tip Sleeve Tip-normal WQP518MA AGND OFFSET_TIP
U7403A ADG5412FBRUZ S1 D1 IN1 OFFSET_TIP OFFSET_PROTECTED C27197a_ENABLE
U7403B ADG5412FBRUZ S2 D2 IN2 AGND
U7403C ADG5412FBRUZ S3 D3 IN3 AGND
U7403D ADG5412FBRUZ S4 D4 IN4 AGND
U7403E ADG5412FBRUZ VDD GND VSS FF +12V −12V AGND
R7409 100000 Ω C27197a_ENABLE AGND
R7410 1000 Ω +5V C27197a_ENABLE
R7411 100000 Ω OFFSET_PROTECTED AGND
R7412 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7402 BAT54S,215 A K COM −12V +12V C360350_SENSE
R7413 100 Ω C8e6645_DRIVE IN_REMOTE
R7414 10000 Ω IN_REMOTE C8e6645_FB
C7403 1e-10 F C8e6645_DRIVE C8e6645_FB
RV1607 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7415 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7416 1e+07 Ω Cc045d2_W_SENSE AGND
R7417 100000 Ω IN_BUFFER ATTEN_SUM
R7418 100000 Ω ATTEN_OUT ATTEN_SUM
C7404 1e-11 F ATTEN_OUT ATTEN_SUM
RV1608 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7419 1000 Ω C38cac0_W C38cac0_W_SENSE
R7420 1e+07 Ω C38cac0_W_SENSE AGND
C7405 1e-08 F C38cac0_W_SENSE AGND
R7421 100 kΩ ATTEN_OUT SUM_NODE
R7422 100 kΩ MANUAL_OFFSET SUM_NODE
R7423 100 kΩ OFFSET_BUFFER SUM_NODE
R7424 100 kΩ SUM_NEG SUM_NODE
R7425 100 kΩ SUM_NEG RESTORE_NODE
R7426 100 kΩ OUT_INTERNAL RESTORE_NODE
R7427 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7428 499 Ω C2073e0_ISO_MID OUT_TIP
R7429 10000 Ω OUT_TIP C2073e0_FB
C7406 1e-10 F C2073e0_DRIVE C2073e0_FB
J7410 Tip Sleeve Tip-normal WQP518MA AGND OUT_TIP
U7404A OPA4197IPWR C534b2a_RAW_5V REF5
U7404B OPA4197IPWR AGND C534b2a_INV REFN5
U7404C OPA4197IPWR C534b2a_GATE_RAW C534b2a_GATE_REF
U7404D OPA4197IPWR C3833e3_SENSE IN_BUFFER
U7404E OPA4197IPWR V+ V− +12V −12V
U7405A OPA4197IPWR C360350_SENSE OFFSET_BUFFER
U7405B OPA4197IPWR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7405C OPA4197IPWR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7405D OPA4197IPWR C38cac0_W_SENSE MANUAL_OFFSET
U7405E OPA4197IPWR V+ V− +12V −12V
U7406A OPA4197IPWR AGND SUM_NODE SUM_NEG
U7406B OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7406C OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7406D OPA4197IPWR AGND PACK_OPA4197IPWR_2_UNUSED3
U7406E OPA4197IPWR V+ V− +12V −12V
C7407 C7408 C7409 C7410 C7411 C7412 C7413 C7414 C7415 C7416 C7417 100 nF +12V −12V AGND
R7430 4990 Ω C6cb7d9_SENSE AGND
D7403 BAS16GW-QX C6cb7d9_DRIVE C6cb7d9_LED_A
D7404 BAS16GW-QX C6cb7d9_SENSE C6cb7d9_LED_A
D7405 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_DRIVE
D7406 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_SENSE
D1608 KT-0603W C6cb7d9_LED_A C6cb7d9_LED_K
R7431 4990 Ω Cd4fa3b_SENSE AGND
D7408 BAS16GW-QX Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7409 BAS16GW-QX Cd4fa3b_SENSE Cd4fa3b_LED_A
D7410 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7411 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_SENSE
D1609 KT-0603W Cd4fa3b_LED_A Cd4fa3b_LED_K
R7432 4990 Ω C3c74b6_SENSE AGND
D7413 BAS16GW-QX C3c74b6_DRIVE C3c74b6_LED_A
D7414 BAS16GW-QX C3c74b6_SENSE C3c74b6_LED_A
D7415 BAS16GW-QX C3c74b6_LED_K C3c74b6_DRIVE
D7416 BAS16GW-QX C3c74b6_LED_K C3c74b6_SENSE
D1610 KT-0603W C3c74b6_LED_A C3c74b6_LED_K
R7433 100000 Ω C75b051_BUF CLIP_HALF
R7434 100000 Ω CLIP_HALF AGND
U7407A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7407B LM393BIDR
U7407C LM393BIDR GND V+ +12V −12V
U7408A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7408B LM393BIDR
U7408C LM393BIDR GND V+ +12V −12V
R7435 22100 Ω +12V C75b051_LED_A
D1610A KT-0603R C75b051_CLIP_N C75b051_LED_A
U7409A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7409B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7409C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7409D OPA4196IDR OUT_INTERNAL C75b051_BUF
U7409E OPA4196IDR V+ V− +12V −12V
C7418 C7419 C7420 C7421 100 nF +12V −12V AGND

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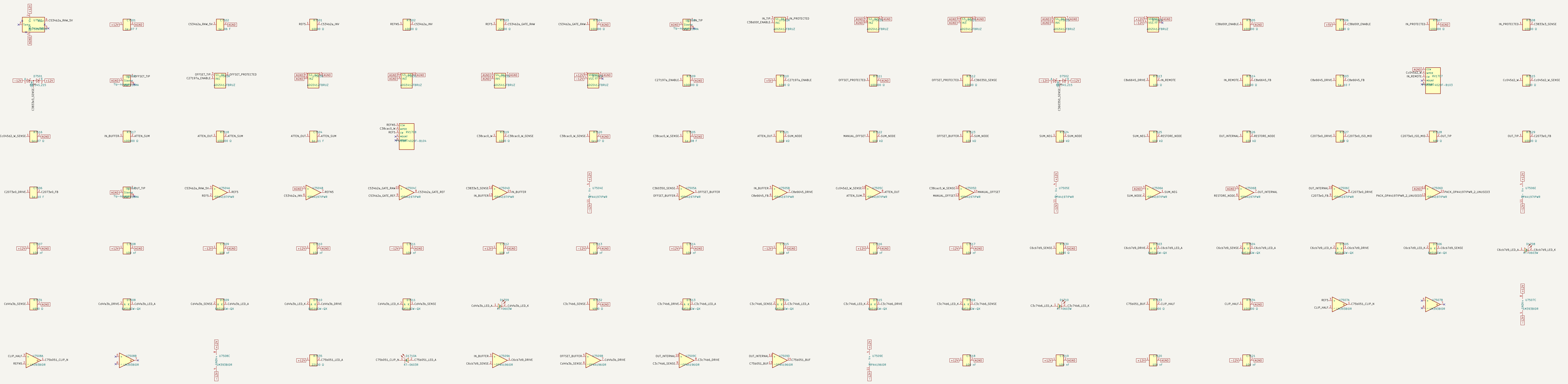
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R7501 REF5 C534b2a_INV 10000 Ω
R7502 REFN5 C534b2a_INV 10000 Ω
R7503 REF5 C534b2a_GATE_RAW 22000 Ω
R7504 C534b2a_GATE_RAW AGND 100000 Ω
J1708 AGND S Tip Sleeve T IN_TIP TN Tip-normals WQP518MA
U7502A IN_TIP S1 D1 IN_PROTECTED C38d00f_ENABLE IN1 ADG5412FBRUZ
U7502B AGND S2 D2 AGND AGND IN2 ADG5412FBRUZ
U7502C AGND S3 D3 AGND AGND IN3 ADG5412FBRUZ
U7502D AGND S4 D4 AGND AGND IN4 ADG5412FBRUZ
U7502E +12V VDD GND AGND -12V VSS FF ADG5412FBRUZ
R7505 C38d00f_ENABLE AGND 100000 Ω
R7506 +5V C38d00f_ENABLE 1000 Ω
R7507 IN_PROTECTED AGND 100000 Ω
R7508 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7501 -12V A K +12V COM C3833e3_SENSE BAT54S,215
J1709 AGND S Tip Sleeve T OFFSET_TIP TN Tip-normals WQP518MA
U7503A OFFSET_TIP S1 D1 OFFSET_PROTECTED C27197a_ENABLE IN1 ADG5412FBRUZ
U7503B AGND S2 D2 AGND AGND IN2 ADG5412FBRUZ
U7503C AGND S3 D3 AGND AGND IN3 ADG5412FBRUZ
U7503D AGND S4 D4 AGND AGND IN4 ADG5412FBRUZ
U7503E +12V VDD GND AGND -12V VSS FF ADG5412FBRUZ
R7509 C27197a_ENABLE AGND 100000 Ω
R7510 +5V C27197a_ENABLE 1000 Ω
R7511 OFFSET_PROTECTED AGND 100000 Ω
R7512 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7502 -12V A K +12V COM C360350_SENSE BAT54S,215
R7513 C8e6645_DRIVE IN_REMOTE 100 Ω
R7514 IN_REMOTE C8e6645_FB 10000 Ω
C7503 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1707 AGND CCW Cc045d2_W WIPER IN_REMOTE CW MOUNT MOUNT PTV09A-4020F-B103
R7515 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7516 Cc045d2_W_SENSE AGND 1e+07 Ω
R7517 IN_BUFFER ATTEN_SUM 100000 Ω
R7518 ATTEN_OUT ATTEN_SUM 100000 Ω
C7504 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1708 REFN5 CCW C38cac0_W WIPER REF5 CW MOUNT MOUNT PTV09A-4020F-B104
R7519 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7520 C38cac0_W_SENSE AGND 1e+07 Ω
C7505 C38cac0_W_SENSE AGND 1e-08 F
R7521 ATTEN_OUT SUM_NODE 100 kΩ
R7522 MANUAL_OFFSET SUM_NODE 100 kΩ
R7523 OFFSET_BUFFER SUM_NODE 100 kΩ
R7524 SUM_NEG SUM_NODE 100 kΩ
R7525 SUM_NEG RESTORE_NODE 100 kΩ
R7526 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7527 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7528 C2073e0_ISO_MID OUT_TIP 499 Ω
R7529 OUT_TIP C2073e0_FB 10000 Ω
C7506 C2073e0_DRIVE C2073e0_FB 1e-10 F
J1710 AGND S Tip Sleeve T OUT_TIP TN Tip-normals WQP518MA
U7504A C534b2a_RAW_5V REF5 REF5 OPA4197IPWR
U7504B AGND C534b2a_INV REFN5 OPA4197IPWR
U7504C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IPWR
U7504D C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7504E +12V V+ V- -12V OPA4197IPWR
U7505A C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7505B IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IPWR
U7505C Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IPWR
U7505D C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IPWR
U7505E +12V V+ V- -12V OPA4197IPWR
U7506A AGND SUM_NODE SUM_NEG OPA4197IPWR
U7506B AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7506C OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7506D AGND PACK_OPA4197IPWR_2_UNUSED3 PACK_OPA4197IPWR_2_UNUSED3 OPA4197IPWR
U7506E +12V V+ V- -12V OPA4197IPWR
C7507 +12V AGND 100 nF
C7508 +12V AGND 100 nF
C7509 -12V AGND 100 nF
C7510 +12V AGND 100 nF
C7511 -12V AGND 100 nF
C7512 +12V AGND 100 nF
C7513 -12V AGND 100 nF
C7514 +12V AGND 100 nF
C7515 -12V AGND 100 nF
C7516 +12V AGND 100 nF
C7517 -12V AGND 100 nF
R7530 C6cb7d9_SENSE AGND 4990 Ω
D7503 C6cb7d9_DRIVE A K C6cb7d9_LED_A BAS16GW-QX
D7504 C6cb7d9_SENSE A K C6cb7d9_LED_A BAS16GW-QX
D7505 C6cb7d9_LED_K A K C6cb7d9_DRIVE BAS16GW-QX
D7506 C6cb7d9_LED_K A K C6cb7d9_SENSE BAS16GW-QX
D1708 C6cb7d9_LED_A C6cb7d9_LED_K KT-0603W
R7531 Cd4fa3b_SENSE AGND 4990 Ω
D7508 Cd4fa3b_DRIVE A K Cd4fa3b_LED_A BAS16GW-QX
D7509 Cd4fa3b_SENSE A K Cd4fa3b_LED_A BAS16GW-QX
D7510 Cd4fa3b_LED_K A K Cd4fa3b_DRIVE BAS16GW-QX
D7511 Cd4fa3b_LED_K A K Cd4fa3b_SENSE BAS16GW-QX
D1709 Cd4fa3b_LED_A Cd4fa3b_LED_K KT-0603W
R7532 C3c74b6_SENSE AGND 4990 Ω
D7513 C3c74b6_DRIVE A K C3c74b6_LED_A BAS16GW-QX
D7514 C3c74b6_SENSE A K C3c74b6_LED_A BAS16GW-QX
D7515 C3c74b6_LED_K A K C3c74b6_DRIVE BAS16GW-QX
D7516 C3c74b6_LED_K A K C3c74b6_SENSE BAS16GW-QX
D1710 C3c74b6_LED_A C3c74b6_LED_K KT-0603W
R7533 C75b051_BUF CLIP_HALF 100000 Ω
R7534 CLIP_HALF AGND 100000 Ω
U7507A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7507B LM393BIDR
U7507C +12V V+ GND -12V LM393BIDR
U7508A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7508B LM393BIDR
U7508C +12V V+ GND -12V LM393BIDR
R7535 +12V C75b051_LED_A 22100 Ω
D1710A C75b051_CLIP_N C75b051_LED_A KT-0603R
U7509A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7509B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7509C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7509D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7509E +12V V+ V- -12V OPA4196IDR
C7518 +12V AGND 100 nF
C7519 +12V AGND 100 nF
C7520 +12V AGND 100 nF
C7521 -12V AGND 100 nF

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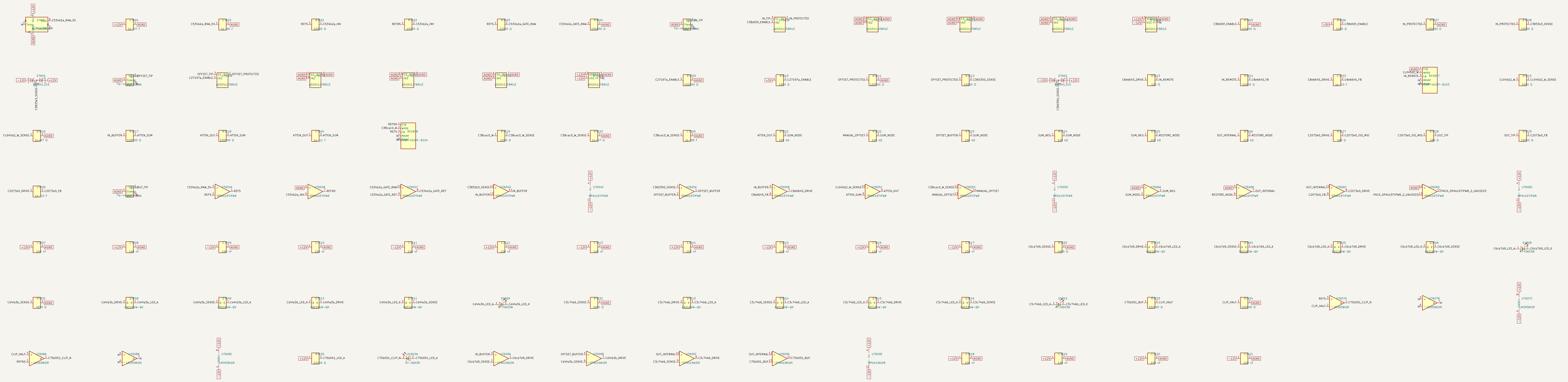
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C7601 +12V AGND 1e-07 F
C7602 C534b2a_RAW_5V AGND 1e-06 F
R7601 REF5 C534b2a_INV 10000 Ω
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R7603 REF5 C534b2a_GATE_RAW 22000 Ω
R7604 C534b2a_GATE_RAW AGND 100000 Ω
J1808 AGND S T TN Tip Sleeve Tip-normal IN_TIP
U7602A IN_TIP C38d00f_ENABLE S1 D1 IN1 IN_PROTECTED ADG5412FBRUZ
U7602B AGND S2 D2 IN2 ADG5412FBRUZ
U7602C AGND S3 D3 IN3 ADG5412FBRUZ
U7602D AGND S4 D4 IN4 ADG5412FBRUZ
U7602E +12V −12V VDD GND VSS FF AGND ADG5412FBRUZ
R7605 C38d00f_ENABLE AGND 100000 Ω
R7606 +5V C38d00f_ENABLE 1000 Ω
R7607 IN_PROTECTED AGND 100000 Ω
R7608 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7601 −12V A K +12V COM BAT54S,215 C3833e3_SENSE
J1809 AGND S T TN Tip Sleeve Tip-normal OFFSET_TIP
U7603A OFFSET_TIP C27197a_ENABLE S1 D1 IN1 OFFSET_PROTECTED ADG5412FBRUZ
U7603B AGND S2 D2 IN2 ADG5412FBRUZ
U7603C AGND S3 D3 IN3 ADG5412FBRUZ
U7603D AGND S4 D4 IN4 ADG5412FBRUZ
U7603E +12V −12V VDD GND VSS FF AGND ADG5412FBRUZ
R7609 C27197a_ENABLE AGND 100000 Ω
R7610 +5V C27197a_ENABLE 1000 Ω
R7611 OFFSET_PROTECTED AGND 100000 Ω
R7612 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7602 −12V A K +12V COM BAT54S,215 C360350_SENSE
R7613 C8e6645_DRIVE IN_REMOTE 100 Ω
R7614 IN_REMOTE C8e6645_FB 10000 Ω
C7603 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1807 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B103
R7615 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7616 Cc045d2_W_SENSE AGND 1e+07 Ω
R7617 IN_BUFFER ATTEN_SUM 100000 Ω
R7618 ATTEN_OUT ATTEN_SUM 100000 Ω
C7604 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1808 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B104
R7619 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7620 C38cac0_W_SENSE AGND 1e+07 Ω
C7605 C38cac0_W_SENSE AGND 1e-08 F
R7621 ATTEN_OUT SUM_NODE 100 kΩ
R7622 MANUAL_OFFSET SUM_NODE 100 kΩ
R7623 OFFSET_BUFFER SUM_NODE 100 kΩ
R7624 SUM_NEG SUM_NODE 100 kΩ
R7625 SUM_NEG RESTORE_NODE 100 kΩ
R7626 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7627 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7628 C2073e0_ISO_MID OUT_TIP 499 Ω
R7629 OUT_TIP C2073e0_FB 10000 Ω
C7606 C2073e0_DRIVE C2073e0_FB 1e-10 F
J1810 AGND S T TN Tip Sleeve Tip-normal OUT_TIP
U7604A C534b2a_RAW_5V REF5 REF5 OPA4197IPWR
U7604B AGND C534b2a_INV REFN5 OPA4197IPWR
U7604C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IPWR
U7604D C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7604E +12V V+ V− −12V OPA4197IPWR
U7605A C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7605B IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IPWR
U7605C Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IPWR
U7605D C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IPWR
U7605E +12V V+ V− −12V OPA4197IPWR
U7606A AGND SUM_NODE SUM_NEG OPA4197IPWR
U7606B AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7606C OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7606D AGND PACK_OPA4197IPWR_2_UNUSED3 PACK_OPA4197IPWR_2_UNUSED3 OPA4197IPWR
U7606E +12V V+ V− −12V OPA4197IPWR
C7607 +12V AGND 100 nF
C7608 +12V AGND 100 nF
C7609 −12V AGND 100 nF
C7610 +12V AGND 100 nF
C7611 −12V AGND 100 nF
C7612 +12V AGND 100 nF
C7613 −12V AGND 100 nF
C7614 +12V AGND 100 nF
C7615 −12V AGND 100 nF
C7616 +12V AGND 100 nF
C7617 −12V AGND 100 nF
R7630 C6cb7d9_SENSE AGND 4990 Ω
D7603 C6cb7d9_DRIVE A K C6cb7d9_LED_A BAS16GW-QX
D7604 C6cb7d9_SENSE A K C6cb7d9_LED_A BAS16GW-QX
D7605 C6cb7d9_LED_K A K C6cb7d9_DRIVE BAS16GW-QX
D7606 C6cb7d9_LED_K A K C6cb7d9_SENSE BAS16GW-QX
D1808 C6cb7d9_LED_A C6cb7d9_LED_K KT-0603W
R7631 Cd4fa3b_SENSE AGND 4990 Ω
D7608 Cd4fa3b_DRIVE A K Cd4fa3b_LED_A BAS16GW-QX
D7609 Cd4fa3b_SENSE A K Cd4fa3b_LED_A BAS16GW-QX
D7610 Cd4fa3b_LED_K A K Cd4fa3b_DRIVE BAS16GW-QX
D7611 Cd4fa3b_LED_K A K Cd4fa3b_SENSE BAS16GW-QX
D1809 Cd4fa3b_LED_A Cd4fa3b_LED_K KT-0603W
R7632 C3c74b6_SENSE AGND 4990 Ω
D7613 C3c74b6_DRIVE A K C3c74b6_LED_A BAS16GW-QX
D7614 C3c74b6_SENSE A K C3c74b6_LED_A BAS16GW-QX
D7615 C3c74b6_LED_K A K C3c74b6_DRIVE BAS16GW-QX
D7616 C3c74b6_LED_K A K C3c74b6_SENSE BAS16GW-QX
D1810 C3c74b6_LED_A C3c74b6_LED_K KT-0603W
R7633 C75b051_BUF CLIP_HALF 100000 Ω
R7634 CLIP_HALF AGND 100000 Ω
U7607A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7607B LM393BIDR
U7607C +12V V+ GND −12V LM393BIDR
U7608A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7608B LM393BIDR
U7608C +12V V+ GND −12V LM393BIDR
R7635 +12V C75b051_LED_A 22100 Ω
D1810A C75b051_CLIP_N C75b051_LED_A KT-0603R
U7609A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7609B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7609C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7609D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7609E +12V V+ V− −12V OPA4196IDR
C7618 +12V AGND 100 nF
C7619 +12V AGND 100 nF
C7620 +12V AGND 100 nF
C7621 −12V AGND 100 nF

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